

Cross-dataset battery life forecasting with time-series foundation models: From zero-shot to prefix-adapted

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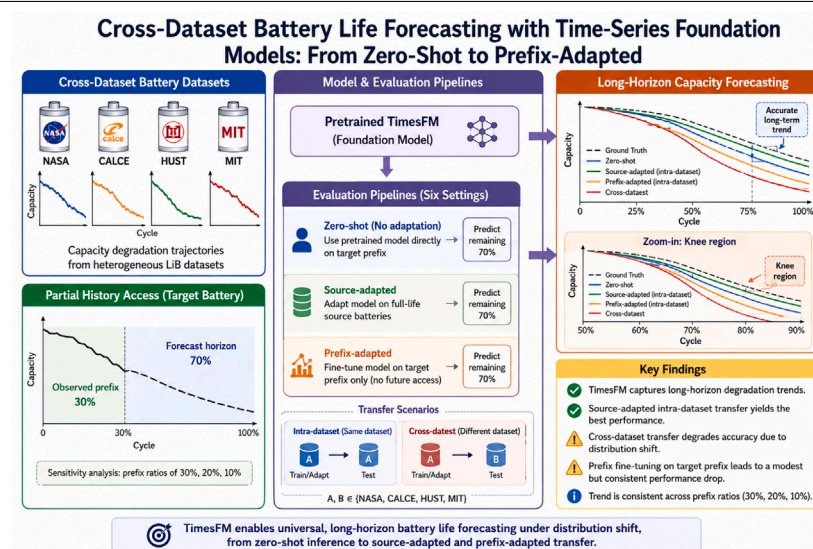
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HIGHLIGHTS

- Capacity-only LiB RUL forecasting using TimesFM foundation models.
- A unified 3x2 protocol evaluates zero-shot and transfer adaptation.
- Cross-dataset forecasting evaluated on four public LiB benchmarks.
- Sensitivity analysis confirms stable trends across prefix ratios.
- Source adaptation helps; cross-dataset transfer remains challenging.

GRAPHICAL ABSTRACT



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ABSTRACT

Reliable remaining useful life estimation is critical for safe and cost-effective battery management, enabling timely maintenance, warranty planning, and preventing unexpected failures in electrified transportation and energy storage systems. However, accurate reliable remaining useful life estimation of lithium-ion batteries is challenging due to distribution shifts across datasets and operating conditions. This work examines whether time-series foundation models can deliver capacity forecasting. We study TimesFM under six evaluation pipelines that span within-dataset and cross-dataset transfer, with and without source/target adaptation. Experiments on four public benchmarks use the first 30% of each target life as history and recursively predict the remaining 70%. Standard accuracy metrics are reported and compared with strong baselines. We further conduct a sensitivity analysis for TimesFM by varying target prefix ratios (30%, 20%, and 10%). Results show that TimesFM captures long-horizon degradation trends; source-adapted transfer yields the best performance when source and target are from the same dataset, while cross-dataset transfer degrades accuracy. Incorporating

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target-prefix fine-tuning yields a modest but consistent drop, and the sensitivity analysis preserves the same trend across split ratios.

1. Introduction

Reliable estimation of the Remaining Useful Life (RUL) of Lithium-ion Batteries (LiBs) is essential for ensuring the safe, reliable, and cost-effective operation of electric vehicles and energy-storage systems [1], as it enables effective maintenance scheduling, risk mitigation, and life-cycle optimization [2]. However, battery degradation arises from complex electrochemical and thermal interactions, making accurate modeling particularly challenging [3].

Existing approaches fall into three groups. Model-based methods rely on a thorough understanding of the physical and electrochemical processes governing battery degradation. These approaches often involve solving coupled differential equations or using empirical formulations [4]. While physically interpretable, such models suffer from high parameter dependency, high computational cost, and poor adaptability to diverse operating conditions and technologies. In contrast, data-driven methods learn degradation patterns directly from data. Techniques ranging from classical machine learning to deep neural networks have shown success in capturing nonlinear degradation behavior [5]. By avoiding explicit physical modeling, these methods offer scalability and flexibility, but typically depend on dataset-specific training and large amounts of labeled data, which limit their generalization to unseen battery types or domains [6]. Recent research has attempted to combine both techniques, forming hybrid frameworks that integrate physical constraints with data-driven inference [7]. Although such models have improved prediction robustness, their generalization capability across datasets remains insufficient, motivating new directions for universal and transferable battery prognostics. In battery prognostics, domain differences are inherently hierarchical. Beyond differences between datasets, substantial shifts may also arise within the same dataset due to variations in charge/discharge protocols, temperature conditions, operating regimes, and dominant ageing mechanisms. Therefore, in this work, cross-dataset transfer is used as a practical operational setting for studying transferability under domain shift, rather than as a complete representation of all possible battery-domain variations.

More recent LiB RUL studies have further extended data-driven prognostics toward interpretable deep generative models, transfer learning under domain shift, and distributed learning frameworks. For example, recent work has explored an interpretable serialized variational autoencoder from a drift-diffusion stochastic equation perspective to improve both interpretability and uncertainty-aware degradation modeling [8]. Other studies have investigated sequential variational autoencoder frameworks with self-attention mechanisms for semi-supervised RUL prediction, highlighting the potential of generative latent-state modeling for complex degradation trajectories [9]. In parallel, transfer learning and domain adaptation strategies have been introduced to address cross-dataset discrepancies and limited target-domain data, including recent studies on battery lifetime prediction under domain shift and source-free RUL prediction [10]. Distributed and privacy-aware learning has also emerged as a relevant direction, with federated learning being explored for LiB RUL prediction under data-isolation constraints [11]. Recent studies have also begun to consider robustness issues associated with imperfect supervision, such as incorrect annotation and label recovery in transfer settings, further highlighting the practical importance of noise-aware learning in real-world prognostic tasks [12]. These advances indicate that recent research is increasingly concerned not only with prediction accuracy, but also with interpretability, transferability, and deployment realism. However, most existing studies still focus on task-specific battery prognostic models, and systematic evidence on the transferability of pretrained time-series foundation models for capacity-trajectory forecasting and downstream RUL estimation remains limited.

Before evaluating advanced learning models, it is essential to clarify how RUL prediction frameworks differ in their input features. Capacity-only approaches rely solely on the capacity fade trajectory, either utilizing the raw capacity series directly [13] or first decomposing it into Intrinsic Mode Functions (IMFs) via Empirical Mode Decomposition (EMD), Variational Mode Decomposition (VMD), or their variants [14], to isolate trend and fluctuation dynamics. These methods are computationally efficient and reproducible across datasets. In contrast, multimodal feature-fusion strategies augment or replace capacity data with voltage, current, or temperature features to capture coupled electrochemical effects and improve early-stage prediction [15]. We recognize the importance of multimodal signals in battery management systems. However, across public datasets, auxiliary sensing modalities often differ in availability, sampling frequency, measurement quality, and preprocessing requirements, which makes controlled cross-dataset comparison more difficult. Therefore, this study adopts a capacity-only setting, not to oppose multimodal modeling, but to provide a unified and reproducible benchmark for evaluating the intrinsic transferability of the forecasting model from the degradation trajectory itself. Extending the present framework to multimodal inputs is an important direction for future work.

The rapid advancement of Generative Artificial Intelligence (Generative AI) has reshaped modern research in machine learning, particularly deep learning. Generative AI refers to a class of algorithms that learn the underlying patterns of complex data and can synthesize realistic new samples across multiple modalities, including text, images, audio, video, and program code [16]. At the core of this paradigm lies the Transformer architecture [17], which supports a broad family of Large Language Models (LLMs) and subsequently inspires the rise of Foundation Models (FMs) across diverse data modalities [18]. Prominent LLM representatives include the GPT series (OpenAI), which use a decoder-only autoregressive Transformer for generative text modeling [19]; BERT (Google), which relies on an encoder-only bidirectional Transformer for masked-token representation learning [20]; and the LLaMA family (Meta), an open-source collection of efficient decoder-only models optimized for scalable pretraining and instruction fine-tuning [21]. These architectures follow a “pre-training-post-adaptation” paradigm: large-scale self-supervised pre-training on diverse corpora, followed by lightweight tuning to adapt to downstream tasks.

Extending this idea to temporal data, time-series FMs adopt the same generative principles but are pre-trained on large and heterogeneous time-series datasets drawn from diverse domains such as finance, climate, healthcare, and energy. Representative examples include TimeGPT [22] (Nixtla Inc.), a decoder-only Transformer trained on large-scale pretraining corpus of sequences; TimesFM [23] (Google Research), a probabilistic decoder-only model enabling long-horizon forecasting; Lag-Llama [24] (Microsoft Research), which applies probabilistic autoregressive decoder-only model with lag/covariate conditioning; Chronos [25] (Amazon AI Labs), a tokenized FM that treats time-series forecasting as language modeling and supports flexible probabilistic inference across diverse frequencies/resolutions; and Moirai [26] (Salesforce Research), a universal probabilistic forecasting FM that integrates numerical and contextual metadata. These time-series FMs inherit the zero-shot reasoning capabilities of LLMs but operate exclusively on numerical temporal data, offering a promising direction for generalizable forecasting and degradation prediction. Recent studies have begun to explore this potential in LiB prognostics. Early work in [27] introduced an LLM-based framework for state-of-health estimation, illustrating the feasibility of transferring generative modeling concepts to battery diagnostics. Building upon this direction, Ref. [28] employed a Lag-Llama-based FM to predict degradation trajectories across connected EV batteries, highlighting the promise

Table 1

Comparison of representative time-series foundation models and justification for the use of TimesFM in this study.

Model family	Core architecture	Forecast style	Pretraining corpus	Context regime
TimesFM	Patched decoder-only transformer	Patch-based autoregressive forecasting	Large real-world and synthetic time-series corpus; public descriptions report about 100 billion real-world time points	Long-context setting; the public TimesFM 2.0 (used in this work) checkpoint supports context lengths up to 2048
TimeGPT	Pre-trained generative transformer for time series	Zero-shot forecasting with optional fine-tuning	Public descriptions emphasize large-scale pretraining over diverse time-series domains, though detailed corpus composition is not fully disclosed	Publicly disclosed context-length details are limited
Lag-Llama	Decoder-only transformer with lag covariates	Probabilistic autoregressive forecasting	Large and diverse time-series corpus	Context length is treated as a tunable hyperparameter
Chronos	Tokenized transformer language-model framework for time series	Scaling and quantization followed by sequence generation	Multi-domain public datasets together with synthetic Gaussian-process series	The original paper uses a context length of 512 during pretraining
Moirai	Masked encoder-based universal forecasting transformer	Universal probabilistic forecasting	Large-scale Open Time Series Archive, with more than 27 billion observations across nine domains	Context length is not tied to a single fixed value; pretraining uses sampled context lengths and supports a relatively flexible context regime

Note: This table is intended to summarize representative time-series foundation model families, rather than to claim that TimesFM is the only valid choice. In this work, TimesFM 2.0 is selected because its public implementation and patch-based forecasting design are well suited to evaluating long-horizon battery degradation forecasting from an early-life prefix.

of few-shot adaptation for cross-dataset RUL forecasting. More recently, authors in [29] proposed a knowledge-distillation strategy that leverages a pre-trained time-series FM to forecast battery capacity degradation, thereby enhancing cross-dataset generalization. Despite recent progress in time-series FMs, their use for RUL prediction of LiBs is still limited, and systematic evidence on cross-dataset generalization is lacking. Therefore, in this work, we evaluate the transferability of TimesFM for LiB capacity-trajectory forecasting. TimesFM is chosen because (i) it is publicly available and reproducible; (ii) it is among the most mature open-source time-series FMs with accessible pretrained weights; and (iii) it operates directly on univariate capacity sequences, making it well suited to capacity-only RUL estimation without multi-modal inputs. To better contextualize this choice, Table 1 compares TimesFM with several representative time-series FMs from the perspectives of core architecture, forecast style, pretraining corpus, and context regime. This comparison is intended to clarify why TimesFM is adopted in this work.

While transfer learning has been widely applied to mitigate data scarcity in battery prognostics, existing approaches can generally be divided into two categories. The first includes conventional transfer-learning frameworks, which train neural networks on a fully observed source domain and then fine-tune them on a partially observed target domain, typically within the same dataset. For example, Ref. [30] analyzed the minimal amount of target data required for effective adaptation. Yet, such models remain limited to intra-dataset transfer and fail to generalize across datasets with different cycling protocols, chemistries, or temperatures. In contrast, FM-based transfer learning represents a new paradigm. Unlike conventional methods, these models encode universal temporal representations from large and heterogeneous pretraining corpora, enabling genuine cross-dataset generalization across distinct datasets and operating conditions. This transition from dataset-specific fine-tuning to FM-driven domain generalization marks a fundamental shift in battery prognostics research and directly motivates the present study.

Building on the above motivation, this study aims to assess whether modern time-series FMs can provide accurate and generalizable RUL prediction for LiBs under distribution shift using capacity-only data. We evaluate TimesFM under six pipelines that factorize *where adaptation happens (none/source-side)* and *whether the target prefix is used for parameter updates*. For fair comparison, we use the first 30% of each target life as historical context and forecast the remainder; Section 4.3 later varies this ratio for sensitivity analysis.

1. **Pretrained, target-free (pure zero-shot):** apply the pretrained model directly to a target cell; no parameter updates on any battery data.
2. **Pretrained, target-adapted (prefix-only):** starting from the pretrained model, perform lightweight adaptation using only the target prefix, then forecast the remaining.
3. **Source-adapted (dataset A), intra-dataset target-free:** adapt the pretrained model on full-life source cells from dataset A, freeze the adapted weights, and evaluate on other targets in dataset A using only their first part prefix.
4. **Source-adapted (dataset A), intra-dataset target-adapted:** further adapt the source-adapted model using only each target's first part prefix (still within dataset A), then forecast the remaining.
5. **Source-adapted (dataset A), cross-dataset target-free:** adapt on full-life source cells from dataset A, freeze the adapted weights, and evaluate on targets from dataset B using only their first part prefix.
6. **Source-adapted (dataset A), cross-dataset target-adapted:** starting from the source-adapted model, perform lightweight adaptation using only the first part prefix of each dataset B target, then forecast the remaining.

Together, these settings span a spectrum from strict target-free zero-shot to lightweight, prefix-only target adaptation, allowing us to quantify how much performance comes from pretraining, from source-side specialization, and from practical per-target tuning under distribution shift.

While the individual ingredients of these pipelines, such as zero-shot inference, source-side adaptation, and prefix-based target adaptation, are rooted in standard practices in transfer learning and time-series forecasting, our goal here is not to claim each component as entirely new in isolation. Instead, we organize them into a unified battery-oriented evaluation framework that disentangles three practically relevant sources of predictive performance: (i) general knowledge inherited from large-scale pretraining, (ii) battery-domain specialization learned from source cells, and (iii) target-specific refinement using only the early-life prefix of a target cell. In the context of battery prognostics, the prefix-only setting is not merely a generic sequence truncation strategy; rather, it corresponds to a realistic early-life deployment scenario in which only the initial portion of a target cell's degradation trajectory is observable before the remaining capacity evolution and RUL must be inferred.

The main contributions of this study are as follows:

- We study a capacity-only setting for LiB RUL prediction to provide a unified and reproducible benchmark for evaluating cross-dataset transferability without the confounding effects of heterogeneous auxiliary sensing modalities.
- We organize a unified battery-oriented 3×2 evaluation framework with six pipelines to disentangle the effects of (i) pre-trained transfer without battery-specific updates, (ii) source-side adaptation using full-life source cells, and (iii) prefix-only target adaptation using only early-life target history while keeping the future strictly unseen. Although these ingredients are individually related to standard transfer-learning and forecasting practice, their integration here is tailored to early-life battery degradation forecasting and downstream RUL estimation under distribution shift.
- We perform a sensitivity analysis of TimesFM across four pipelines by varying prefix ratios (30%/70%, 20%/80%, 10%/90%), and show that the qualitative trends are stable under reduced history.
- We conduct cross-dataset transfer experiments on four public datasets to evaluate transferability under dataset-level domain shift.
- We provide reproducible implementation details and competitive baselines to contextualize when FMs help and where they fail under long-horizon recursive forecasting.

The remainder of this paper is organized as follows. Section 2 reviews TimesFM. Section 3 describes the experimental framework and dataset configuration. Section 4 reports and analyzes results. Finally, Section 5 concludes the paper and discusses future research directions.

2. TimesFM: A decoder-only foundation model for time-series

TimesFM is a decoder-only Transformer trained with causal self-attention and a patching scheme. In this scheme, L_{in} and L_{out} denote the *input-patch length* and *output-patch length* (in time-points), respectively. Importantly, L_{in} is the length of a *single* input patch, while the model can condition on multiple input patches. Hence, the effective context length can be $k \times L_{in}$ for $k \geq 1$.

Training example: Consider a time series of length 512 with $L_{in} = 32$ and $L_{out} = 128$. A single series yields multiple supervised training pairs by using progressively longer prefixes (multiple input patches) to predict a fixed-length future patch. Specifically, the model is trained to (i) use points 1–32 to forecast points 33–160, (ii) use points 1–64 (two input patches) to forecast points 65–192, (iii) use points 1–96 (three input patches) to forecast points 97–224, and so on. In each pair, all L_{out} predicted points contribute to the training loss. The construction of training prefix-to-future pairs under the patching scheme is illustrated in Fig. 1.

Inference example: During inference, if the required forecast horizon exceeds L_{out} , the model generates future patches recursively. For instance, given an observed prefix of length 256 (i.e., eight input patches when $L_{in} = 32$) and tasked with forecasting the next 256 points, the model first generates points 257–384 (one output patch of length 128), then conditions on the observed prefix plus the generated outputs to generate points 385–512 (the next output patch). Therefore, a larger L_{out} reduces the number of recursive generation steps (e.g., $256/128 = 2$ steps) compared with using $L_{out} = L_{in} = 32$ (which would require $256/32 = 8$ steps), thereby mitigating error accumulation in long-horizon forecasting. To visualize the long-horizon rollout, Fig. 2 compares recursive inference when $L_{out} = 128$ versus $L_{out} = 32$, highlighting the resulting 2-step vs. 8-step generation.

2.1. Architecture

TimesFM is a decoder-only Transformer trained with causal self-attention for autoregressive time-series forecasting [23]. The input numeric series is first partitioned into contiguous (non-overlapping)

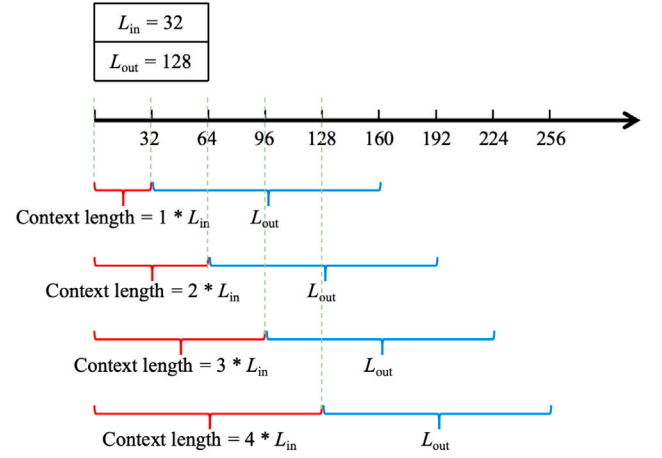


Fig. 1. Construction of training pairs under TimesFM patching (take $L_{in} = 32$ and $L_{out} = 128$ for example).

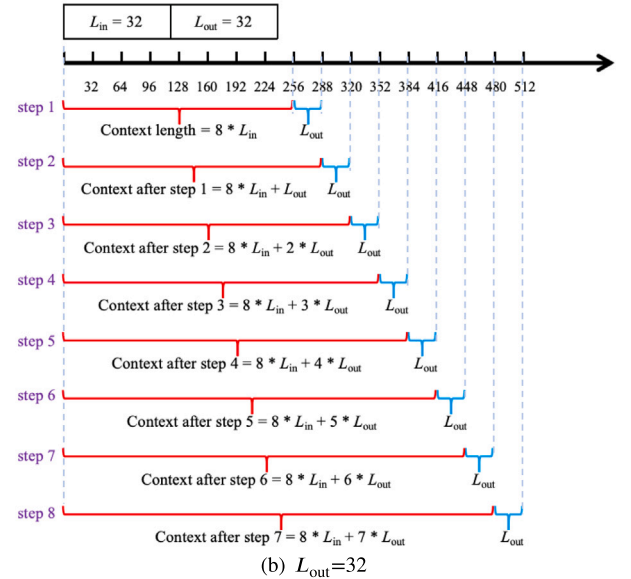
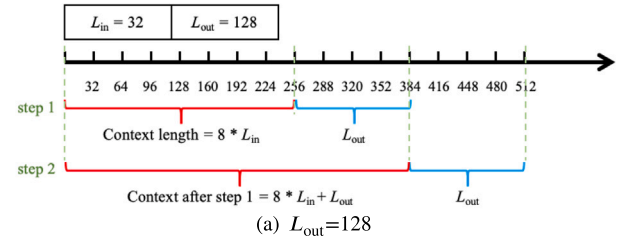


Fig. 2. Recursive inference with different output-patch lengths (take $L_{in} = 32$ and $L_{out} = 128/32$ for example).

patches of length L_{in} , and each patch is projected to the model dimension to form tokens. These tokens are then processed by a stack of Transformer layers, each consisting of causal Self-Attention (SA) and a position-wise Feed-Forward Network (FFN) with residual connections. Following the original TimesFM design, we keep layer normalization inside the Transformer layers, but do not add additional normalization in the residual blocks [23]. The model produces a future patch of length L_{out} per generation step, and longer horizons are obtained by recursively rolling out future patches.

Table 2

Composition of TimesFM pretraining dataset.

Source: Adapted from [23].

Dataset	Granularity	# Time series	# Time points
Synthetic	–	3,000,000	6,144,000,000
Electricity	Hourly	321	8,443,584
Traffic	Hourly	862	15,122,928
Weather	10 Min	42	2,213,232
Favorita Sales	Daily	111,840	139,179,538
LibCity	15 Min	6,159	34,253,622
M4 hourly	Hourly	414	353,500
M4 daily	Daily	4,227	9,964,658
M4 monthly	Monthly	48,000	10,382,411
M4 quarterly	Quarterly	24,000	2,214,108
M4 yearly	Yearly	22,739	840,644
Wiki hourly	Hourly	5,608,693	239,110,787,496
Wiki daily	Daily	68,448,204	115,143,501,240
Wiki weekly	Weekly	66,579,850	16,414,251,948
Wiki monthly	Monthly	63,151,306	3,789,760,907
Trends hourly	Hourly	22,435	393,043,680
Trends daily	Daily	22,435	122,921,365
Trends weekly	Weekly	22,435	16,585,438
Trends monthly	Monthly	22,435	3,821,760

2.2. Pretraining corpus, leakage risk, and transfer rationale

TimesFM is pretrained on a large mixture of real-world and synthetic time-series to cover diverse domains, temporal granularities, and trend/seasonality patterns. According to the original TimesFM report, the pretraining corpus is dominated by three major sources: (i) Wikipedia Pageviews, (ii) Google Trends, and (iii) synthetic time-series generated from statistical pattern generators (e.g., Auto-Regressive Moving Average (ARMA), seasonal mixtures, trend and step functions). In addition, several public forecasting benchmarks (e.g., M4 and traffic/electricity/weather-related datasets) are included as auxiliary real-world sources. The reported dataset mixing strategy samples 80% real data and 20% synthetic data, and uses different maximum context lengths depending on the granularity (e.g., 512 for sufficiently long series, 256 for weekly, and 64 for $\geq$ monthly data), together with a reversible instance normalization variant applied on the context statistics. Table 2 reports the main components of the TimesFM pretraining corpus, including the data source, temporal granularity, and the resulting scale in terms of the number of series and time points.

Composition of TimesFM pretraining dataset is directly relevant to the potential data-leakage concern. Based on the original TimesFM report, the publicly documented pretraining corpus consists mainly of large-scale web and synthetic time series, together with several standard public forecasting datasets (M4, Electricity, Traffic, and Weather). We do not find NASA, CALCE, or the other LiB degradation benchmarks used in this work among these explicitly documented pretraining components. Therefore, to the best of our knowledge, the evaluated battery datasets are not part of the publicly described TimesFM pretraining mixture. Nevertheless, since the full underlying training corpus has not been exhaustively released, we avoid making an absolute claim of zero overlap. Instead, we conservatively treat TimesFM as a black-box pretrained model and adopt strictly separated downstream evaluation protocols to mitigate any potential leakage effects.

Beyond the data-leakage issue, the substantial gap between the pretraining source domains and LiB degradation data raises a second question: why should representations learned from such general time series transfer to battery capacity forecasting at all? Although the source domains used to pretrain TimesFM, such as web search and page-view time series, are clearly different from the electrochemical processes underlying LiB degradation, the intended transfer mechanism in this work is not based on shared physical semantics. Instead, a plausible explanation is that large-scale pretraining helps the model

Table 3

Representative TimesFM configurations and the checkpoint used in this work.

Size	Layers	Model dim	L_{in}	L_{out}	Heads
500M (used)	50	1280	32	128	16
200M	20	1280	32	128	16
70M	10	1024	32	128	16
17M	10	512	32	128	16

• Our experiments use the TimesFM 2.0 PyTorch checkpoint `google/timesfm-2.0-500m-pytorch`.

• The 200M/70M/17M entries are representative configurations reported in [23].

learn domain-agnostic temporal priors and forecasting operators, such as long-term trend continuation, local fluctuation modeling, change-point sensitivity, and autoregressive sequence extension under variable context lengths. This interpretation is supported by the published TimesFM design, whose pretraining corpus combines real-world and synthetic time series, with the synthetic component explicitly including piece-wise linear trends, ARMA processes, seasonal patterns, and step functions. For battery capacity forecasting, the most relevant bridge is therefore not semantic similarity between source and target domains, but the fact that battery degradation trajectories also exhibit structured temporal dynamics, including smooth decline, local perturbations, and possible regime transitions. Moreover, the patched decoder-only architecture predicts future patches from observed historical patches, which aligns naturally with the present prefix-to-suffix battery forecasting setting. From this perspective, TimesFM is not assumed to encode battery-specific electrochemistry a priori; rather, it provides a pre-trained forecasting prior whose usefulness for LiB capacity-trajectory forecasting and downstream RUL estimation can then be tested and adapted in a downstream setting.

2.3. Model sizes and checkpoint configuration

The public TimesFM configurations specify the number of layers, model width, attention heads, and patch lengths. Table 3 summarizes representative settings reported in [23]. In this study, we use the publicly available TimesFM 2.0 PyTorch checkpoint `google/timesfm-2.0-500m-pytorch`.

2.4. Version comparison and rationale for using TimesFM 2.0

TimesFM 2.5 reduces the number of parameters to about 200M, extends the supported context length to 16k, and provides an optional quantile head (about 30M parameters) for continuous quantile forecasts up to 1k steps. It also removes the explicit frequency indicator and adds several forecasting flags. Because a stable PyTorch release and weights aligned with our pipeline are not generally available for 2.5 at the time of our study, we report results with the vetted 2.0 weights to ensure reproducibility.

3. Experimental framework

3.1. Dataset description

Four public datasets are adopted to evaluate the proposal. For each one, only a subset of cells is selected to ensure cross-dataset diversity [14,31,32]:

- NASA (18650 LiB Dataset, nominal capacity 2.0 Ah): cells B0005/B0006/B0007 are tested at 24 °C with Constant Current (CC) – Constant Voltage (CV) charging (1.5A to 4.2V, CV hold to 20 mA) and 2.0A discharging to cell-specific cutoffs (B0005: 2.7 V; B0006: 2.5 V; B0007: 2.2 V); termination at 1.4 Ah threshold.

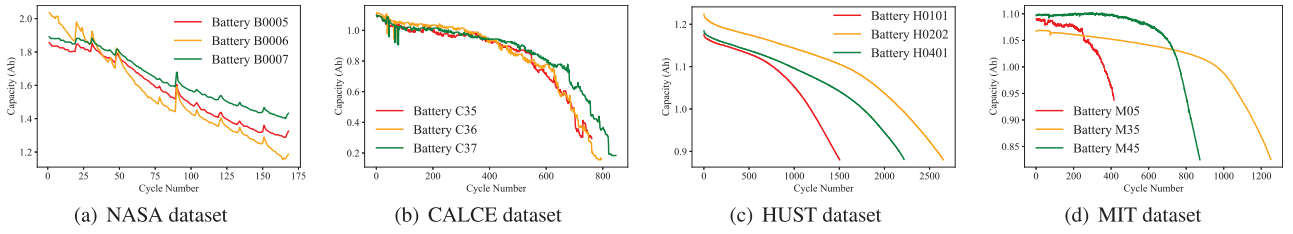


Fig. 3. Capacity degradation trajectories of batteries from four datasets (NASA/CALCE/HUST/MIT).

- CALCE (Sony 18650 LiB Dataset, nominal capacity 1.1 Ah): cells CS2-35, CS2-36, and CS2-37 (abbreviated as C35/C36/C37) follow CC–CV charging to 4.2 V (taper to 20 mA), discharge to 2.7 V, and a termination at 0.77 Ah threshold.
- HUST (LFP/Graphite LiB Dataset, nominal capacity 1.1 Ah, 3.3 V): This dataset contains LFP/Graphite cells cycled at a constant temperature of 30 °C using an identical fast-charging protocol but distinct multi-stage discharge protocols. In this work, three cells (H0101, H0202, and H0401) are selected, each corresponding to a different discharge protocol group, to evaluate the model's adaptability under diverse discharge-rate conditions.
- MIT (LFP/Graphite LiB Dataset, nominal capacity 1.1 Ah): This study utilizes the third batch of the MIT fast-charging dataset, containing commercial LFP/Graphite cells cycled to failure at 30 °C under one-step or two-step fast-charging protocols. Three cells (M05, M35, and M45) are selected for analysis to cover diverse charging strategies:

- M05 follows 4.65C(69%) – 6C;
- M35 follows 5.6C(26%) – 4.5C;
- and M45 follows 5.2C(58%) – 4C.

In the two-step fast-charging notation $C_1(p\%)$ – C_2 , the percentage $p\%$ denotes the SOC switching point between the two constant-current (CC) stages: the cell is first charged at rate C_1 from 0– $p\%$ SOC, and then at rate C_2 from $p\%$ –80% SOC. All protocols end the second CC step at 80% SOC, after which the cells undergo a 1C CC–CV charging phase to 3.6 V, followed by CC discharging at 4C to 2.0 V.

The corresponding capacity curves are presented in Fig. 3. Across the four benchmarks, both lifespan and degradation shape vary markedly. NASA cells show relatively short lives with steady decline; CALCE cells run much longer and exhibit a late, sharp knee to End of Life (EOL); HUST cells have the longest horizons with smooth, gradual fade; MIT cells mix short-life and long-life behaviors. These differences highlight the cross-dataset variability in cycle counts and capacity trajectories.

3.2. Evaluation metrics

Several metrics are employed to comprehensively assess the prediction performance. Let $\mathbf{y} = \{y_m\}_{m=1}^M$ denote the true capacity values and $\hat{\mathbf{y}} = \{\hat{y}_m\}_{m=1}^M$ the corresponding predictions, where M is the number of test samples. Specifically, Normalized Root Mean Square Error (NRMSE), Normalized Mean Absolute Error (NMAE), Mean Absolute Percentage Error (MAPE), and R^2 are defined in (1)–(4), respectively [14].

$$\text{NRMSE} = \sqrt{\frac{1}{M} \sum_{m=1}^M (y_m - \hat{y}_m)^2} / y_0 \quad (1)$$

$$\text{NMAE} = \left(\frac{1}{M} \sum_{m=1}^M |y_m - \hat{y}_m| \right) / y_0 \quad (2)$$

$$\text{MAPE}(\%) = \frac{100}{M} \sum_{m=1}^M \left| \frac{y_m - \hat{y}_m}{y_m} \right| \quad (3)$$

$$R^2 = 1 - \frac{\sum_{m=1}^M (y_m - \hat{y}_m)^2}{\sum_{m=1}^M (y_m - \bar{y})^2} \quad (4)$$

where, NRMSE, which is normalized by the initial capacity y_0 , penalizes large point-wise deviations and reflects overall trajectory accuracy on a scale-free basis. NMAE normalizes the absolute error by the initial capacity y_0 , enabling fair comparisons across batteries/datasets with different capacity scales. MAPE reports a widely used percentage error and is easy to interpret in practice. R^2 measures goodness-of-fit and how well the degradation trend is captured.

3.3. Experimental pipelines and data access policy

Fig. 4 summarizes the six experimental pipelines. All pipelines operate on capacity-only inputs and evaluate forecasts using the metrics defined in Section 3.2. To avoid ambiguity, we explicitly specify the data-access policy used throughout this work in Table 4.

Overall, the six pipelines are designed to isolate complementary effects:

- Pretraining transfer: compare pure zero-shot and prefix-only adaptation starting from the same pretrained checkpoint (P1 vs. P2).
- Source-side adaptation value: test whether full-life source cells provide useful priors that improve forecasting on unseen targets (P1/P2 vs. P3–P6).
- Cross-dataset transfer under shift: P5/P6 adapt on dataset A but are evaluated on dataset B, thereby explicitly probing robustness under distribution shift.
- Target-prefix adaptation under shift: quantify the marginal benefit of adapting on only the target prefix within the same dataset and across datasets (P3 vs. P4; P5 vs. P6).
- Strict data-access control: in all cases, only the first percentage (30% for example) of each target life is observable at adaptation time, and no target future points are used for training, preventing leakage and enabling fair comparisons.

To make the six pipelines (P1–P6) fully reproducible, Table 5 lists the exact source (adaptation) and target (test) cells used for each dataset. P1–P2 are target-only evaluations (no source cells are used), thus only the test targets are specified. For P3–P4 (intra-dataset), we adapt on one full-life source cell within each dataset and evaluate on two remaining target cells from the same dataset. For P5–P6 (cross-dataset), the source adaptation uses the combined source set (NASA-B0005 and CALCE-C35). Evaluation is then performed on all target cells in the rightmost column, including seen-like targets (from NASA/CALCE, i.e., same datasets as the sources and thus distribution-closer) and unseen targets (from HUST/MIT, i.e., from different datasets and thus distribution-shifted).

3.4. Baselines

We evaluate six pipelines (P1–P6) that differ only in (i) whether source full-life cells are used for adaptation, (ii) whether the target prefix is allowed to update model parameters, and (iii) whether evaluation

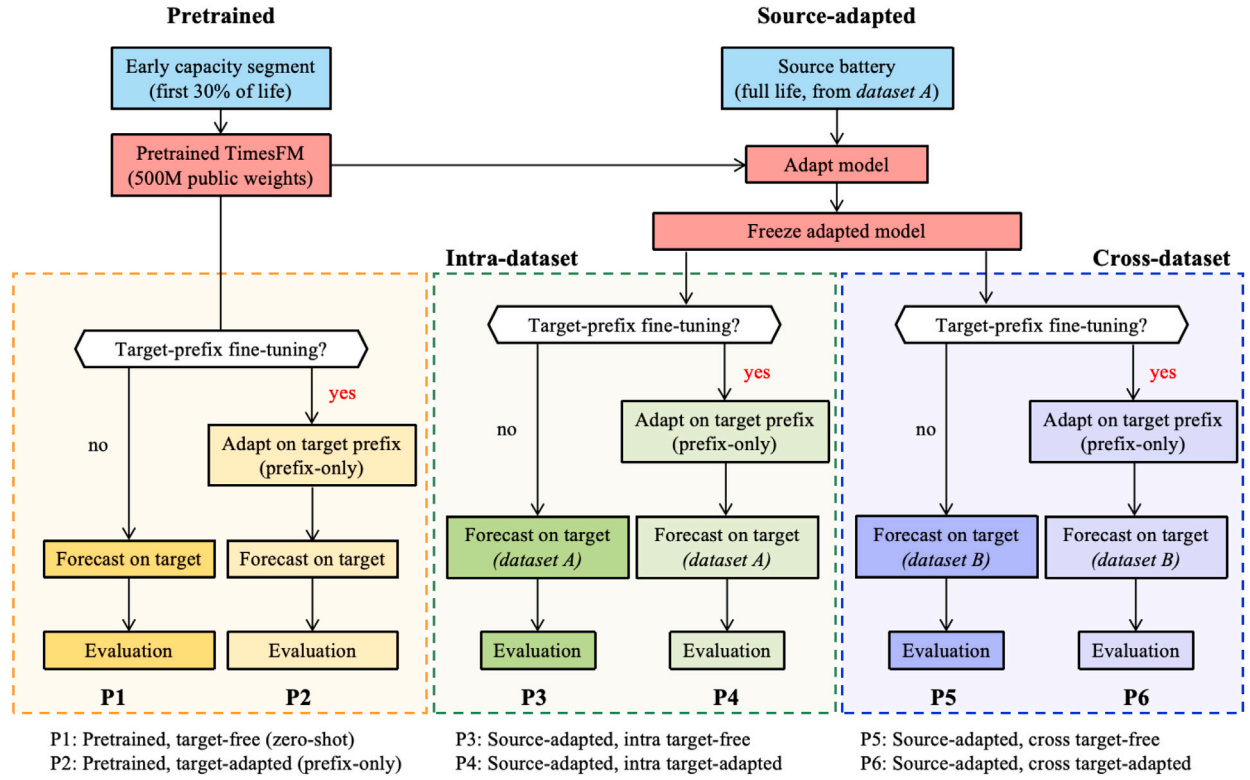


Fig. 4. Overview of the six experimental pipelines (P1–P6) in this work: P1–P2 denote pretrained forecasting with and without target-prefix adaptation, P3–P4 denote source-adapted intra-dataset transfer with and without target-prefix adaptation, and P5–P6 denote source-adapted cross-dataset transfer with and without target-prefix adaptation.

Table 4

Summary of the six experimental pipelines (P1–P6) used in this work, including their transfer setting, source-data access policy, target-domain usage, and whether target-prefix fine-tuning or target-future information is allowed.

ID	Pipeline name	Source for adaptation?	Target	Target prefix fine-tune?	Target future used?
P1	Pretrained, target-free (zero-shot)	No	A/B	No	No
P2	Pretrained, target-adapted (prefix-only)	No	A/B	Yes (prefix only)	No
P3	Source-adapted (A), intra target-free	Yes (full-life)	A	No	No
P4	Source-adapted (A), intra target-adapted	Yes (full-life)	A	Yes (prefix only)	No
P5	Source-adapted (A), cross target-free	Yes (full-life)	B	No	No
P6	Source-adapted (A), cross target-adapted	Yes (full-life)	B	Yes (prefix only)	No

Table 5

Train/test cell selection under the six pipelines (P1–P6), showing for each setting whether source cells are used for adaptation and which target cells are evaluated in the pretrained, intra-dataset, and cross-dataset transfer scenarios.

Pipeline	Dataset	Train (source) cells	Test (target) cells
P1/P2	NASA	–	B0006, B0007
	CALCE	–	C36, C37
	HUST	–	H0202, H0401
	MIT	–	M35, M45
P3/P4	NASA	B0005	B0006, B0007
	CALCE	C35	C36, C37
	HUST	H0101	H0202, H0401
	MIT	M05	M35, M45
P5/P6	NASA: B0005		B0006, B0007
	CALCE: C35		C36, C37 H0202, H0401 M35, M45

- P1/P2 use no source cells.
- P5/P6 perform joint source adaptation on NASA-B0005 and CALCE-C35 and then evaluate on all target cells listed in the rightmost column.

is intra-dataset (dataset A) or cross-dataset (dataset B). To ensure fair comparisons under different data-access policies, we use two groups of baselines:

- **Group I baselines** for P1–P2, contextualizing the performance of pretrained TimesFM under strict target-only access. These baselines require no source training data and operate using only the target prefix, matching the data access of P1–P2. We include:
 - TimeGPT (API-based pretrained FM);
 - AutoReg (autoregressive linear model fitted on the target prefix);
 - Exponential decay curve fitting on the target prefix;
 - and Isotonic regression applied to prefix-fitted predictions.
- **Group II:** trainable deep sequence baselines for P3–P6. Pipelines P3–P6 explicitly involve adaptation using full-life source cells (and optionally the target prefix). Therefore, we compare against standard trainable forecasters that can be trained on the same available data under each pipeline’s policy. We include LSTM, GRU, TCN, and a standard Transformer forecaster, trained on the source set (and, when permitted, further updated on the target

prefix). This group aims to evaluate whether TimesFM's source-side adaptation offers advantages over conventional sequence models.

For compact reporting in figures and tables, we denote each method as **P#-Method**, where **P#** identifies the pipeline (P1–P6) and **Method** is one of: TimesFM, TimeGPT, AutoReg, ExpDecay, Isotonic, LSTM, GRU, TCN, Transformer. For example, P1-TimesFM denotes pretrained target-free TimesFM, while P6-Transformer denotes a Transformer trained with source adaptation and further updated on the target prefix under the cross-dataset setting.

3.5. Implementation details and configuration

Reproducible benchmark configurations and hyperparameter search spaces are reported in [Tables 6–8](#).

- TimeGPT is accessed through the Nixtla API. Since no public weights are provided and the API does not support weight fine-tuning, we only report target-free (P1) results for TimeGPT.
- AutoReg, ExpDecay, and Isotonic are target-only baselines: they fit on the target prefix and extrapolate to the horizon. Thus, they are reported only under P2 (prefix-only) and do not apply to P1 (target-free).
- P4 and P6 are two-stage pipelines: their Stage-1 is identical to P3 and P5 (source-adapted), and Stage-2 additionally applies prefix-only target finetuning.
- LSTM, GRU, TCN, and Transformer are trainable backbones used in P3–P6. P3/P5 search model-specific and shared training hyperparameters, while P4/P6 freeze the backbone and fine-tune only the prediction head on the target prefix.
- Whenever a validation split is used on the target prefix ([Table 6](#): P1/P2-TimesFM and P2-Isotonic; [Table 7](#): LSTM/GRU/TCN/Transformer; [Table 8](#): TimesFM), we split the target prefix into 80%:20% train/validation for hyperparameter selection and early stopping.
- Optuna objective is to minimize NRMSE.
- Post-processing (monotone projection, ceil/floor, normalization) is applied when enabled to enforce non-increasing and bounded predictions. To preserve battery-specific physical consistency, monotonic projection together with upper- and lower-bound constraints is applied, so that the predicted capacity remains non-increasing over cycles, does not exceed historical capacity values, and stays non-negative.
- TimesFM uses a fixed patch length (32), so context length is rounded to multiples of 32 and clamped to a minimum of 32. In contrast, LSTM/GRU/TCN/Transformer have no such patch constraint and can use 16.

For clarity, the values $L_{in} = 32$ and $L_{out} = 128$ shown in [Figs. 1–2](#) refer to the published TimesFM checkpoint configuration rather than the unique downstream forecasting setting used in this work. In our experiments, the actual context length is selected from {32, 64} and the forecast horizon from {16, 32, 64}. This distinction is important because it separates the checkpoint's illustrative patch design from the actual rollout settings used in the battery forecasting experiments. In addition, the implementation does not require the sequence length to be divisible by the forecast horizon: during rolling inference, the model predicts one block at a time, and for the final incomplete block only the required remaining steps are retained.

3.6. Experimental environment

All experiments are executed on a single NVIDIA RTX A6000 GPU (48 GB VRAM) with NVIDIA driver 550.127.08 and CUDA 12.4, under a Python 3.11 conda environment with PyTorch 2.5.1. Experiments are launched from a remote Visual Studio Code session. TimesFM uses the public 2.0 weights via a PyTorch implementation (checkpoint: google/timesfm-2.0-500m-pytorch).

4. Results and analysis

To ensure fair comparisons under the different data-access policies of P1–P6, we present baselines in two groups: Group I consists of methods matching P1–P2, while Group II includes trainable deep forecasters trained on source data (and optionally updated on the target prefix when allowed) to match the adaptation setting of P3–P6.

4.1. Group I results

Prediction results and evaluation metrics are shown in [Figs. 5 and 6](#), respectively. In the heatmaps, the best-performing method for each battery (per metric) is highlighted by a black circle around the corresponding value. The inference time is reported in [Table 9](#).

Overall, prefix adaptation is the key driver of accuracy: (1) The P1-TimesFM often produces overly smooth (sometimes near-flat) extrapolations after the split and fails to capture the post-knee acceleration. (2) In contrast, P2-TimesFM achieves the best or near-best performance on most batteries. (3) P1-TimeGPT is competitive and attains the best overall metrics on NASA B0007, suggesting that some targets are already well-covered by the pretrained FM prior. (4) On CALCE C37, all methods struggle, and the P2-Isotonic baseline becomes the strongest among the simple prefix-only methods, indicating that for highly irregular/noisy degradation, enforcing a conservative monotone fit plus extrapolation can be more robust than aggressively adapting TimesFM. (5) Regarding efficiency, classical prefix-only baselines (AutoReg/ExpDecay/Isotonic) are essentially instantaneous, whereas P1-TimeGPT and P1/P2-TimesFM incur a clear compute overhead.

Quantitatively, P2-TimesFM is the strongest overall method in Group I. Across the eight target batteries, it achieves the best mean values on all four metrics, with NRMSE = 0.0476, NMAE = 0.0360, MAPE = 6.0397, and $R^2 = 0.7676$. Compared with the second-best method, this corresponds to a 34.9% reduction in mean NRMSE (vs. P2-Isotonic, 0.0730), a 28.3% reduction in mean NMAE (vs. P1-TimeGPT, 0.0502), and a 35.7% reduction in mean MAPE (vs. P2-Isotonic, 9.3918), while exceeding the second-best mean R^2 by 0.3868 (vs. P1-TimeGPT, 0.3808). In addition, P2-TimesFM ranks first on 6 of the 8 target batteries across all four metrics, confirming that prefix adaptation is the main driver of the Group I accuracy gain for TimesFM.

4.2. Group II results

For the P3–P6 pipelines, capacity forecasts are compared in [Fig. 7](#), while the battery-wise metrics and distributions are summarized in [Figs. 8 and 9](#), respectively. In the heatmaps, the circled value marks the best method within each pipeline for a given battery, while boldface highlights the best performance across pipelines for that battery. The boxplots summarize the distribution of each metric across the eight batteries for every method–pipeline combination (P3–P6 × LSTM/GRU/TCN/Transformer/TimesFM). The median is shown by the central green line, the box spans the interquartile range, and whiskers indicate the non-outlier range.

Quantitatively, P3-TimesFM is the strongest overall setting, achieving the best mean values across the eight target batteries, with NRMSE = 0.0336, NMAE = 0.0267, MAPE = 4.4682, and $R^2 = 0.8961$. P5-TimesFM ranks second, with mean NRMSE = 0.0429 and mean $R^2 = 0.8053$. By contrast, P4-TimesFM degrades to mean NRMSE = 0.0586 and mean $R^2 = 0.6947$, while P6-TimesFM performs worst, with mean NRMSE = 0.0942, and mean $R^2 = 0.1416$. Relative to P3, P6 increases mean NRMSE by 180.2%, while reducing mean R^2 by 0.7545. These results show that intra-dataset source adaptation without target-prefix fine-tuning is the most reliable transfer setting, whereas cross-dataset target-prefix adaptation is the most difficult. The CALCE cells remain the main source of difficulty. For example, under P4-TimesFM, the mean NRMSE on CALCE is 0.1746, compared

Table 6

Hyperparameter search spaces for the Group I baselines used in the pretrained and target-prefix-adapted settings (P1–P2).

Model	Setting
P1-TimeGPT	model: <code>timegpt-1</code> (Nixtla TimeGPT-1)
P1-TimesFM	Optuna search with 24 trials: context length $\in \{32, 64\}$; horizon length $\in \{16, 32, 64\}$; batch size $\in \{16, 32, 64, 128\}$ best trial selected by the optimization metric
P2-TimesFM	Fix the best P1-TimesFM (context length/horizon length/batch size), then run Optuna search with 50 trials over fine-tuning learning rate $\in [1e-6, 5e-5]$ (log-uniform), LR scheduler settings (patience $\in \{10, \dots, 15\}$, decay factor $\in [0.6, 0.9]$) LoRA is applied to all target modules: lora rank $\in \{8, 16\}$, lora alpha $\in \{16, 32\}$
P2-AutoReg	Fit an $AR(p)$ model on the target prefix: $y_t = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t$ α : constant (bias) term; ϕ_i : coefficient for the value observed (i) time steps before the current time $p \in \{30, 31, \dots, 100\}$, step = 1, selected by minimizing the Bayesian Information Criterion (BIC) on the target prefix
P2-ExpDecay	Fit $\hat{y}(t) = \hat{a}e^{-\hat{b}t} + \hat{c}$ on the target prefix for each $\hat{c}$ estimate $(\hat{a}, \hat{b})$ by regression of $\log(y_t - \hat{c}) = \log \hat{a} - \hat{b}t$ (invalid if $y_t \leq \hat{c}$) grid-search $\hat{c} \in [r_{\min} y_{\min}, r_{\max} y_{\min}]$, $y_{\min} = \min_{t \leq c} y_t$ (prefix minimum), ($r_{\min} = 0$, $r_{\max} = 0.95$, grid = 100)
P2-Isotonic	Fit a decreasing isotonic regression on the target prefix via Pool-Adjacent-Violators Algorithm (PAVA) $\hat{z} = \arg \min_z \sum_{t \leq c} (y_t - z_t)^2$, s.t. $z_t \geq z_{t+1}$, $\forall t < c$; z_t the monotone fitted sequence; $\hat{z}_t$ the isotonic solution on the prefix $\hat{y}_t := \hat{z}_t$ ($t \leq c$), $\hat{y}_{c+h} = \hat{z}_c + s h$ ($h \geq 1$); h the forecast step; s the extrapolation slope estimated $s(w) = \frac{1}{w} \sum_{i=1}^w (\hat{z}_{c-i+1} - \hat{z}_{c-i})$; enforce $s(w) \leq 0$ (decreasing); $w \in \{1, 2, \dots, 20\}$, step = 1, selected on prefix validation

- y_t denotes the observed capacity, while $\hat{y}$ (and $\hat{z}$) denote fitted/predicted values.

- c is the cut index (prefix length), h is the forecast step.

- The values $L_{\text{in}} = 32$ and $L_{\text{out}} = 128$ in Figs. 1–2 illustrate the published patching configuration of the TimesFM checkpoint. In our downstream experiments, however, the actual context length and forecast horizon are tuned separately, with context length selected from $\{32, 64\}$ and forecast horizon selected from $\{16, 32, 64\}$.

Table 7

Hyperparameter search spaces for the trainable backbone models used in the source-adapted pipelines (P3–P6).

Model	P3/P5 (Model-specific)	P3/P5 (Shared)	P4/P6 (Shared)
LSTM	hidden size $\in \{32, 64, 128, 256\}$ number of layers $\in \{1, 2, 3\}$	context length $\in \{16, 32, 64\}$ horizon length $\in \{16, 32, 64\}$	
GRU	hidden size $\in \{32, 64, 128, 256\}$ number of layers $\in \{1, 2, 3\}$	batch size $\in \{16, 32, 64, 128\}$ dropout $\in [0.0, 0.3]$	fine-tuning the output prediction head all backbone layers are frozen learning rate = $0.1 \times lr$ epochs = 200 early-stopping patience = 30 early stopping min delta = $1e-4$ stride = 1
TCN	channels $\in \{32, 64, 128, 256\}$ number of layers $\in \{1, 2, 3\}$ kernel size $\in \{2, 3, 5\}$	learning rate $lr \in [1e-4, 5e-3]$ LR patience $\in \{10, \dots, 15\}$ LR decay factor $\in [0.6, 0.9]$	
Transformer	$d_{\text{model}} \in \{8, 16, 32, 64, 128\}$ $\leq \in \{2, 4, 8\}$ $d_{\text{model mod}} \leq 0$ number of layers $\in \{1, 2, 3, 4\}$ $d_{\text{FFN}} \in \{64, 128, 256, 512, 1024\}$	data augmentation probability $\in [0, 0.3]$ time-warp magnitude $\in [0, 0.1]$ knee-shift magnitude $\in [0, 0.1]$ noise standard deviation 0.002 scaling magnitude 0.005	

- noise standard deviation = 0.002 and scaling magnitude = 0.005 (applied when augmentation is enabled).

Table 8

TimesFM-specific hyperparameter settings used in the source-adapted pipelines (P3–P6).

Hyperparameter	P3	P4	P5	P6
Specific	/	Learning rate = $0.1 \times lr$ augmentation is off	/	Learning rate = $0.1 \times lr$ augmentation is off
Shared	context length $\in \{32, 64\}$; horizon length $\in \{16, 32, 64\}$; batch size $\in \{16, 32, 64, 128\}$ learning rate $lr \in [1e-5, 5e-4]$; LR patience $\in \{10, \dots, 15\}$; LR decay factor $\in [0.6, 0.9]$ epochs = 200; weight decay = $1e-5$; early stop patience = 25; early stop min delta = $1e-4$ LoRA on (rank $\{8, 16\}$, alpha $\{16, 32\}$) data augmentation on (probability $[0, 0.3]$, time-warp $[0, 0.1]$, knee-shift $[0, 0.1]$, standard deviation 0.002, scal 0.005)			

- The values $L_{\text{in}} = 32$ and $L_{\text{out}} = 128$ in Figs. 1–2 illustrate the published patching configuration of the TimesFM checkpoint. In our downstream experiments, however, the actual context length and forecast horizon are tuned separately, with context length selected from $\{32, 64\}$ and forecast horizon selected from $\{16, 32, 64\}$.

Table 9

Inference time (s) of Group I baselines (P1–P2) for each target battery under the pretrained and target-prefix-adapted settings (P1–P2).

Battery ID	P1-TimeGPT	P1-TimesFM	P2-TimesFM	P2-AutoReg	P2-ExpDecay	P2-Isotonic
NASA B0006	13.12	4.23	53.97	0.01	0.01	0.00
NASA B0007	13.25	5.68	35.28	0.01	0.01	0.00
CALCE C36	73.29	30.34	184.75	1.06	0.02	0.01
CALCE C37	72.40	7.14	205.08	0.12	0.01	0.01
HUST H0202	430.18	22.95	785.94	0.28	0.02	0.02
HUST H0401	360.74	17.39	67.61	0.33	0.02	0.02
MIT M35	98.74	6.16	191.01	2.17	0.02	0.01
MIT M45	73.81	4.34	55.47	0.13	0.02	0.01

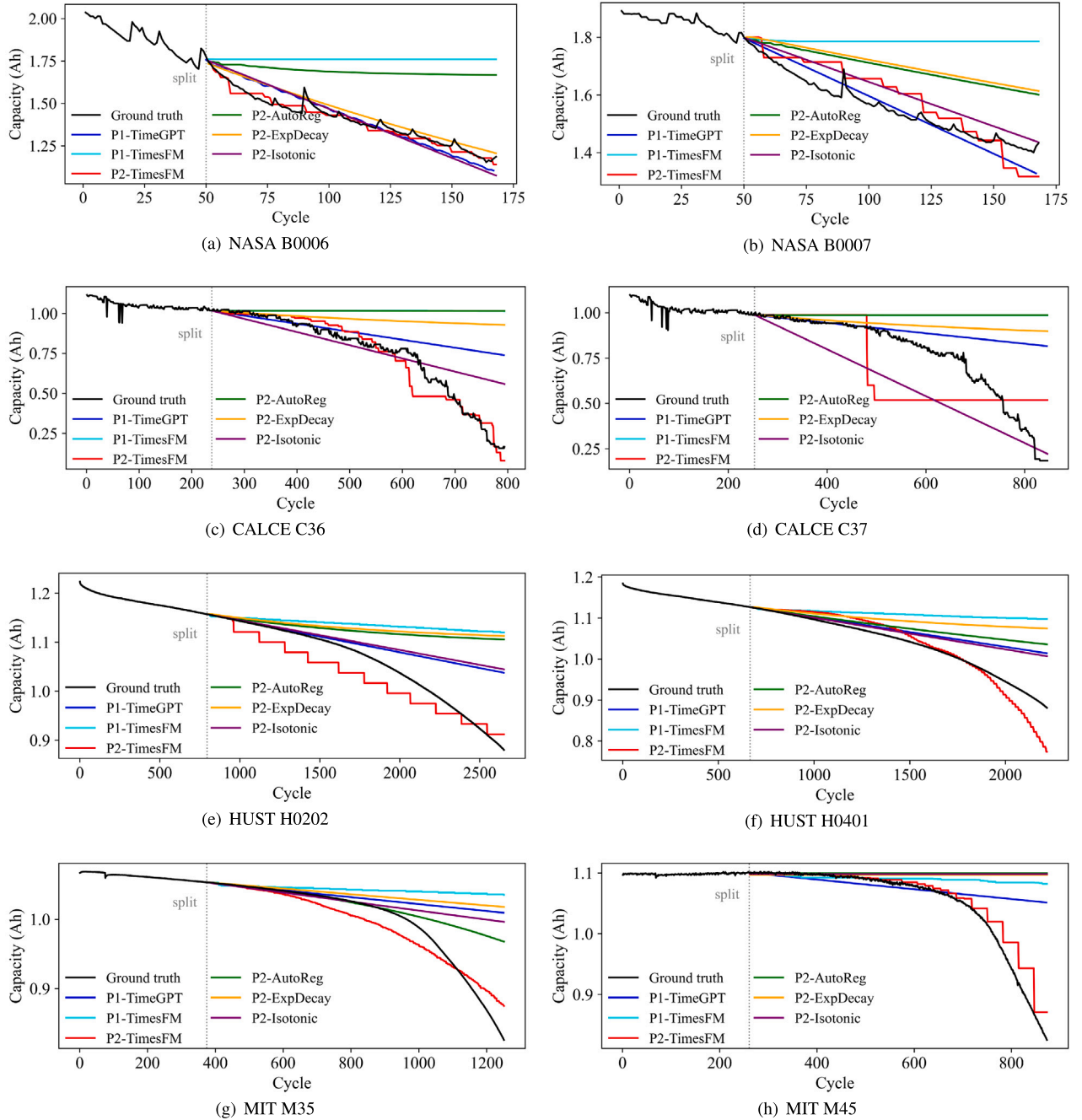


Fig. 5. Capacity trajectory forecasts of the target batteries under the pretrained and target-prefix-adapted Group I methods (P1–P2).

with 0.0199 on the remaining datasets; under P6-TimesFM, the corresponding values are 0.2178 and 0.0530. This confirms that CALCE contributes disproportionately to the overall degradation in transfer performance.

It can be concluded that: (1) intra-dataset source adaptation without target-prefix fine-tuning (P3) most consistently yields faithful post-split trajectories, especially around the knee region and at long horizons. (2) In contrast, cross-dataset adaptation with prefix fine-tuning (P6) shows larger performance dispersion and occasional R^2 degradation for lots of method–battery pairs, indicating transfer mismatch across domains. (3) Comparing target-free (P3/P5) vs. target-adapted (P4/P6) within each domain, P4 and P6 are generally not beneficial for the trainable baselines LSTM/GRU/TCN/Transformer, as prefix-only fine-tuning (head-only or LoRA on selected target modules) is driven by

a short or smooth prefix and thus overfits local statistics. This re-weighting improves the fit to the prefix but reduces sensitivity to non-stationary dynamics and knee points after the split. The adapted models become overly smooth and less responsive in long-horizon forecasts, which widens the error distribution and lowers R^2 . Consequently, P4/P6 often degrade performance relative to the target-free P3/P5 counterparts. (4) Although using the target prefix for adaptation can hurt final accuracy due to overfitting local prefix statistics, P4/P6-TimesFM remains beneficial relative to classic baselines (P4/P6-LSTM/GRU/TCN/Transformer) because the adapted FM starts from a stronger pretrained representation. Even with a short or stable prefix, the pretrained model retains broader temporal priors and domain-level patterns, so its forecasts degrade less and generalize better than fully-trained small models. (5) TimesFM performs relatively modestly on CALCE, likely because the capacity trajectories exhibit frequent small

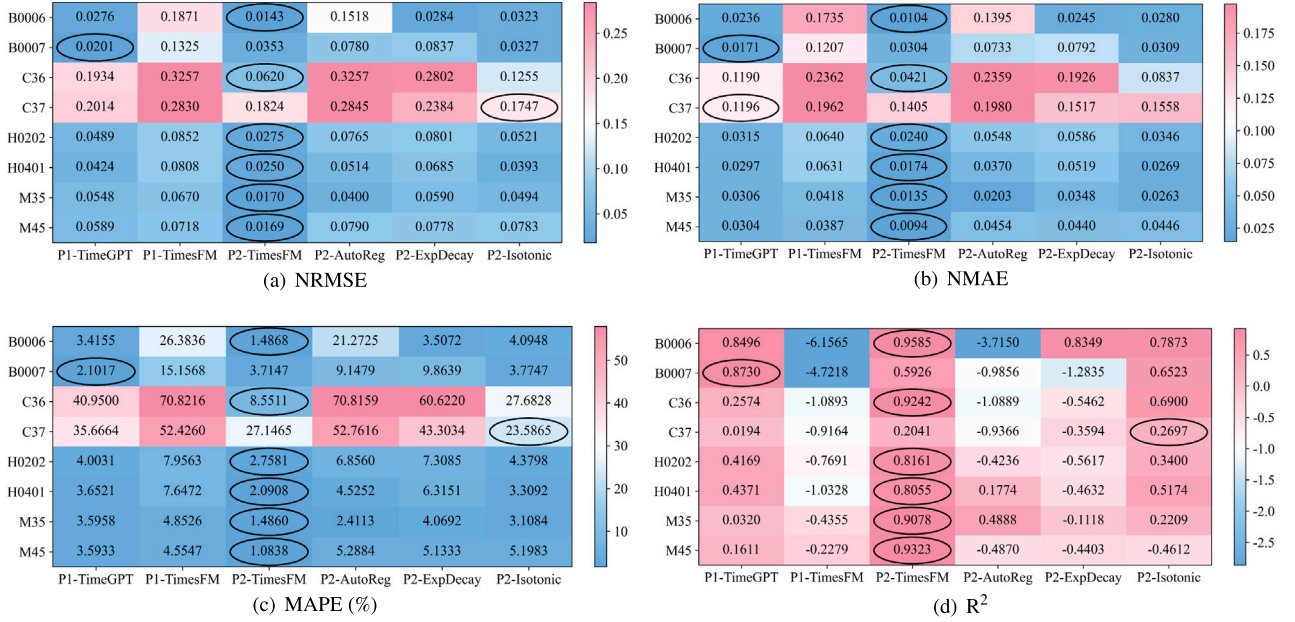


Fig. 6. Evaluation metrics of Group I baselines (P1–P2) under the pretrained and target-prefix-adapted settings (P1–P2).

Table 10

Training time (s) of Group II baselines during Stage-1 source-side adaptation for P3 and P5.

ID	Method	NASA	CALCE	HUST	MIT
P3	LSTM	3.79	8.02	9.68	1.98
	GRU	4.05	3.17	10.94	1.18
	TCN	2.50	4.14	8.71	1.83
	Transformer	3.26	10.01	11.64	2.76
	TimesFM	71.48	188.20	145.26	233.42
P5	LSTM			7.29	
	GRU			5.01	
	TCN			3.54	
	Transformer			11.43	
	TimesFM			148.27	

fluctuations and local perturbations, which weaken the model's ability to stably fit long-term degradation trends.

Tables 10–12 summarize the computational cost of Group II baselines. Table 10 reports stage-1 training time: P3 measures intra-dataset source adaptation, while P5 measures cross-dataset source adaptation. Table 11 isolates the stage-2 target-prefix fine-tuning time (P4/P6), making explicit that these values are additional adaptation costs rather than full retraining. Table 12 reports inference time for each stage (P3–P6), enabling a direct comparison of test-time latency across pipelines and methods.

It can be concluded that: TimesFM has substantially higher stage-1 training cost (P3/P5) than the classical baselines, reflecting its larger model size. However, the stage-2 target-prefix fine-tuning time (P4/P6) and inference time are relatively moderate, making TimesFM computationally heavy mainly during source adaptation rather than during target-side refinement or deployment.

4.3. Sensitivity analysis

All prior analyses used the first 30% of the target series as the prefix to forecast the remaining 70%, under which TimesFM achieved solid and stable performance across the pipelines, except for CALCE. Compared with the other datasets, the CALCE cells considered here are more difficult to forecast under the present prefix-based protocol because they combine stronger local irregular fluctuations, relatively

long recursively predicted suffixes under the same split ratio, and less predictable late-stage degradation transitions. To assess sensitivity to the amount of available history, we therefore shorten the prefix and evaluate how TimesFM's accuracy evolves across P3–P6 on the remaining datasets. In addition, NASA sequences are relatively short, and TimesFM requires a minimum context length of 32; therefore, for NASA we only evaluate 30% and 20% as history, while for HUST and MIT, we evaluate all three splits (30%/70%, 20%/80%, and 10%/90%).

Fig. 10 shows TimesFM prediction trajectories under different target-prefix ratios, while Fig. 11 reports averaged metrics over four batteries (HUST and MIT) across the available pipelines for each split. The qualitative trend is consistent with prior findings: source-adapted transfer performs best when source and target are from the same dataset, cross-dataset transfer degrades accuracy, and target-prefix fine-tuning yields a consistent drop across splits.

To further examine the robustness of the proposed evaluation setting, we additionally tested more target cells from HUST and MIT datasets for the P3–P4 pipelines and more source–target combinations for the P5–P6 pipelines. These supplementary results are reported in Appendices A and B, respectively, and show conclusions consistent with the main text.

We also examine error accumulation during recursive inference using HUST and MIT datasets as illustrative long-horizon cases. They are selected because they have the longest cycle lives among the batteries, making them particularly informative for examining whether recursive errors grow too rapidly over extended forecasting horizons. Fig. 12 reports the step-wise recursive NRMSE under three prefix ratios (target prefix 30%/20%/10%) and four settings (P3–P6). Here, the horizontal axis denotes the recursive step index, i.e., the rollout block number during recursive forecasting; for a target battery with suffix length H and selected forecast horizon h , the total number of recursive steps is $S = \lceil H/h \rceil$, where h is selected from {16, 32, 64} by the hyperparameter search.

Across all four cells, the recursive NRMSE exhibits stage-dependent rather than uniformly explosive growth. Several conclusions can be drawn: (i) In P3, the error remains low for most recursive steps and increases only gradually, indicating that source-adapted target-free forecasting is the most stable setting under long recursive rollout. (ii) P5 shows a broadly similar behavior, although moderate local fluctuations appear in several cases. (iii) By contrast, P4 and P6 are

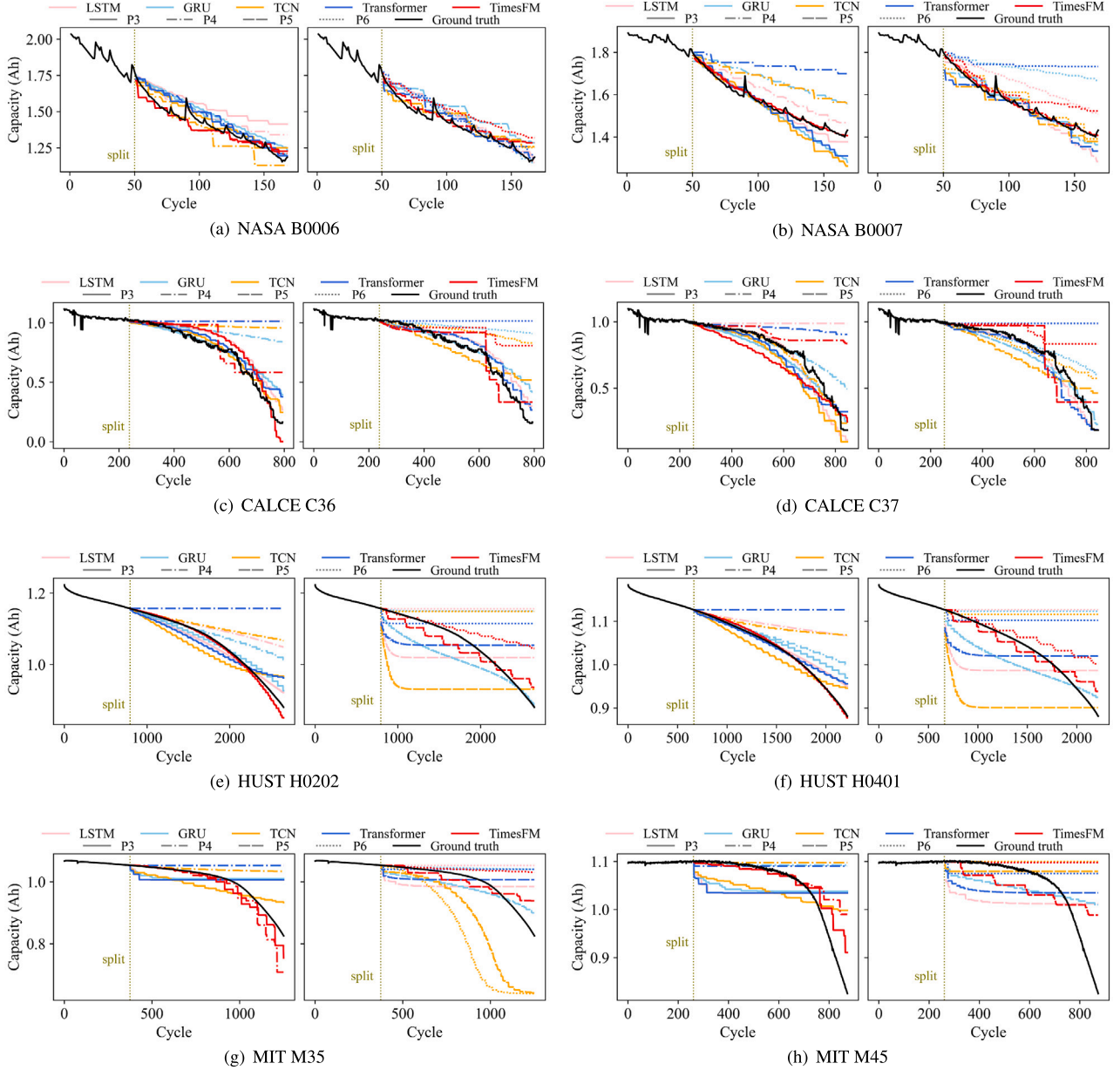


Fig. 7. Capacity trajectory forecasts of the target batteries under the source-adapted Group II methods in the intra-dataset and cross-dataset settings (P3–P6).

Table 11

Training time (s) of Group II baselines during Stage-2 target-prefix fine-tuning on each target battery in P4 and P6.

Pipeline	Method	B0006	B0007	C36	C37	H0202	H0401	M35	M45
P4 (only report stage 2)	LSTM	1.26	0.69	2.15	2.74	2.18	2.90	4.45	2.00
	GRU	1.05	0.66	3.59	1.56	2.24	2.26	1.23	1.12
	TCN	0.67	0.61	4.95	1.86	3.85	6.63	2.71	2.51
	Transformer	1.37	0.76	2.39	2.70	4.83	2.26	1.46	3.32
	TimesFM	8.08	13.63	15.12	18.93	1.00	1.00	23.15	18.99
P6 (only report stage 2)	LSTM	0.87	0.82	1.44	1.46	5.69	4.42	5.28	2.17
	GRU	0.76	0.74	0.97	1.51	6.25	7.46	3.78	3.81
	TCN	1.65	0.91	0.99	0.66	2.30	1.29	1.38	1.13
	Transformer	1.03	0.73	3.56	4.10	5.49	5.01	3.06	2.37
	TimesFM	24.17	15.61	17.84	20.02	25.35	26.37	16.59	12.04

	NRMSE																			
	0.05 0.10 0.15 0.20 0.25																			
B0006	0.0707	0.0414	0.0261	0.0299	0.0228	0.0442	0.0301	0.0358	0.0344	0.0226	0.0348	0.0540	0.0279	0.0249	0.0229	0.0317	0.0325	0.0267	0.0338	0.0493
B0007	0.0143	0.0282	0.0352	0.0254	0.0082	0.0310	0.0643	0.0615	0.1055	0.0073	0.0243	0.0181	0.0209	0.0251	0.0077	0.0529	0.0968	0.0189	0.1099	0.0344
C36	0.0894	0.0873	0.0469	0.0768	0.0761	0.3264	0.2446	0.2935	0.3237	0.1359	0.0773	0.0977	0.1102	0.0609	0.0874	0.3264	0.2775	0.2422	0.3264	0.2289
C37	0.0665	0.0584	0.1080	0.0773	0.1026	0.2845	0.0959	0.0334	0.2487	0.2132	0.0443	0.0578	0.1134	0.0728	0.1095	0.2845	0.1265	0.1090	0.2845	0.2066
H0202	0.0139	0.0156	0.0316	0.0233	0.0076	0.0551	0.0422	0.0611	0.1030	0.0076	0.0679	0.0423	0.1152	0.0616	0.0228	0.1026	0.0986	0.0979	0.0791	0.0527
H0401	0.0187	0.0271	0.0194	0.0183	0.0031	0.0657	0.0374	0.0639	0.0959	0.0031	0.0648	0.0375	0.1178	0.0543	0.0176	0.0959	0.0937	0.0889	0.0804	0.0384
M35	0.0555	0.0558	0.0370	0.0558	0.0212	0.0759	0.0756	0.0654	0.0759	0.0359	0.0548	0.0306	0.1351	0.0556	0.0296	0.0759	0.0691	0.1848	0.0683	0.0648
M45	0.0647	0.0626	0.0507	0.0649	0.0272	0.0781	0.0754	0.0780	0.0745	0.0430	0.0691	0.0519	0.0700	0.0635	0.0454	0.0784	0.0790	0.0791	0.0686	0.0782
	NMAE																			
	0.05 0.10 0.15																			
B0006	0.0663	0.0386	0.0213	0.0251	0.0175	0.0400	0.0251	0.0301	0.0297	0.0171	0.0294	0.0504	0.0241	0.0199	0.0178	0.0254	0.0264	0.0230	0.0272	0.0469
B0007	0.0116	0.0217	0.0265	0.0195	0.0062	0.0296	0.0608	0.0573	0.0965	0.0055	0.0185	0.0145	0.0173	0.0186	0.0058	0.0503	0.0890	0.0156	0.0992	0.0293
C36	0.0735	0.0532	0.0338	0.0531	0.0633	0.2366	0.1647	0.2035	0.2333	0.0957	0.0617	0.0593	0.0832	0.0511	0.0715	0.2366	0.1938	0.1659	0.2366	0.1639
C37	0.0420	0.0499	0.0843	0.0563	0.0899	0.1980	0.0572	0.0234	0.1673	0.1314	0.0312	0.0504	0.0973	0.0505	0.0848	0.1980	0.0731	0.0678	0.1980	0.1376
H0202	0.0121	0.0115	0.0278	0.0194	0.0049	0.0373	0.0274	0.0414	0.0807	0.0049	0.0610	0.0373	0.1021	0.0531	0.0199	0.0801	0.0753	0.0744	0.0573	0.0344
H0401	0.0112	0.0181	0.0167	0.0110	0.0025	0.0494	0.0251	0.0459	0.0774	0.0026	0.0577	0.0344	0.1063	0.0475	0.0135	0.0774	0.0747	0.0690	0.0606	0.0266
M35	0.0407	0.0391	0.0330	0.0409	0.0141	0.0513	0.0508	0.0399	0.0513	0.0190	0.0470	0.0286	0.1033	0.0407	0.0206	0.0513	0.0440	0.1468	0.0425	0.0399
M45	0.0537	0.0499	0.0426	0.0539	0.0154	0.0443	0.0416	0.0442	0.0416	0.0221	0.0628	0.0408	0.0408	0.0520	0.0349	0.0446	0.0454	0.0455	0.0411	0.0443
	MAPE (%)																			
	10 20 30 40 50																			
B0006	10.0069	5.6261	3.0831	3.5591	2.5248	6.0239	3.5669	4.5168	4.2545	2.4374	4.1808	7.4710	3.5773	2.8151	2.7147	3.5813	3.7388	3.4007	3.8394	7.0049
B0007	1.4506	2.7569	3.3840	2.4690	0.7411	3.6456	7.5743	7.1537	12.1117	0.6554	2.3200	1.7589	2.0692	2.2530	0.6963	6.2382	11.1523	1.8920	12.4753	3.6927
C36	17.9699	18.5890	9.9163	16.9025	13.6189	70.9714	52.8312	63.5103	70.3275	29.0016	16.3293	20.6870	24.7058	15.8245	12.7727	70.9714	60.2091	52.4169	70.9714	60.2755
C37	9.6487	8.6348	16.3985	11.3105	14.5713	52.7616	17.3689	4.8186	45.7638	38.6175	5.8936	8.5617	19.8678	9.3090	16.9098	52.7616	22.6211	20.1028	52.7616	38.4699
H0202	1.4257	1.4221	3.2768	2.3265	0.6146	4.7097	3.4760	5.2233	9.9682	0.6153	7.0346	1.8440	11.4249	6.3266	2.3382	9.8976	9.3410	9.2318	7.1459	4.3655
H0401	1.3923	2.2349	1.9603	1.3604	0.2854	6.0138	3.0982	5.6224	9.3387	0.2916	6.5151	3.8795	11.8553	5.5059	1.6138	9.3387	9.0270	8.3695	7.3707	3.2681
M35	4.6043	4.4566	3.6036	4.6273	1.5960	5.8952	5.8441	4.6387	5.8952	2.2305	5.1736	3.0856	11.5797	4.6001	2.3403	5.8952	5.0934	16.2198	4.9325	4.6408
M45	5.8592	5.4935	4.6503	5.8807	1.7941	5.1695	4.8749	5.1552	4.8674	2.6082	6.7144	4.5075	4.7401	5.6921	3.8668	5.2035	5.2929	5.3032	4.7488	5.1702
	R ²																			
	-1.5 -1.0 -0.5 0.0 0.5																			
B0006	-0.0217	0.6489	0.8605	0.8166	0.8940	0.6011	0.8148	0.7375	0.7573	0.8957	0.7519	0.4043	0.8413	0.8732	0.8931	0.7951	0.7839	0.8545	0.7657	0.5032
B0007	0.9330	0.7404	0.5955	0.7894	0.9781	0.6869	-0.3489	-0.2321	-2.6265	0.9819	0.8079	0.8936	0.8574	0.7947	0.9808	0.0861	-2.0567	0.8832	-2.9397	0.6150
C36	0.8427	0.8498	0.9568	0.8837	0.8859	-1.0980	-0.1783	-0.6968	-1.0639	0.6362	0.8823	0.8120	0.7607	0.9271	0.8496	-1.0980	-0.5166	-0.1551	-1.0980	-0.0323
C37	0.8943	0.9185	0.7207	0.8569	0.7482	-0.9366	0.7799	0.9733	-0.4796	-0.0875	0.9531	0.9202	0.6926	0.8732	0.7131	-0.9366	0.6171	0.7157	-0.9366	-0.0213
H0202	0.9531	0.9409	0.7571	0.8682	0.9860	0.2595	0.5656	0.0916	-1.5858	0.9860	-0.1235	0.5635	-2.2332	0.0747	0.8738	-1.5620	-1.3676	-1.3354	-0.5223	0.3243
H0401	0.8916	0.7703	0.8878	0.8956	0.9970	-0.3468	0.5630	-0.2722	-1.8677	0.9969	-0.3105	0.5622	-3.3237	0.0806	0.9032	-1.8677	-1.7365	-1.4623	-1.0166	0.5392
M35	0.0149	0.0049	0.5620	0.0062	0.8558	-0.8413	-0.8248	-0.3657	-0.8413	0.5885	0.0402	0.7003	-4.8341	0.0104	0.7206	-0.8413	-0.5272	-9.9120	-0.4908	-0.3404
M45	0.0026	0.0667	0.3884	-0.0031	0.8237	-0.4527	-0.3531	-0.4493	-0.3210	0.5601	-0.1358	0.3573	-0.1686	0.0391	0.5082	-0.4642	-0.4882	-0.4908	-0.1212	-0.4549
P3-LSTM P3-GRU P3-TCN P3-Transformer P3-TimesFM P4-LSTM P4-GRU P4-TCN P4-Transformer P4-TimesFM P5-LSTM P5-GRU P5-TCN P5-Transformer P5-TimesFM P6-LSTM P6-GRU P6-TCN P6-Transformer P6-TimesFM																				

Fig. 8. Evaluation metrics of the Group II baselines under the source-adapted intra-dataset and cross-dataset settings (P3-P6).

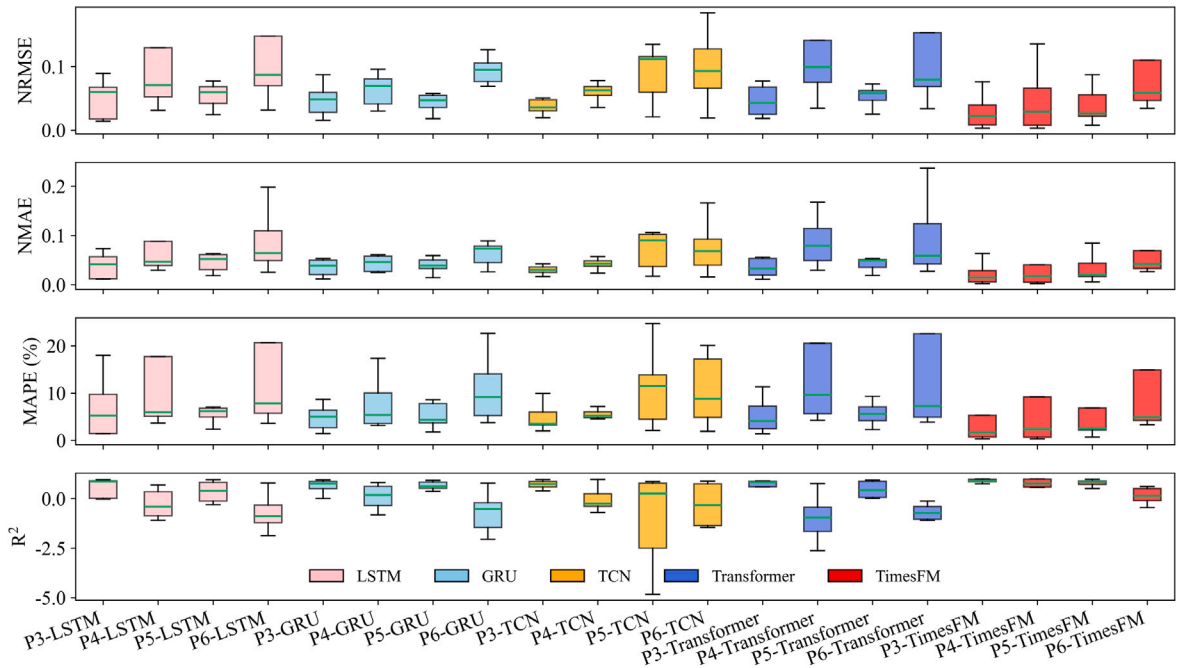


Fig. 9. Battery-wise metric distributions of the Group II baselines under the source-adapted intra-dataset and cross-dataset settings (P3-P6). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

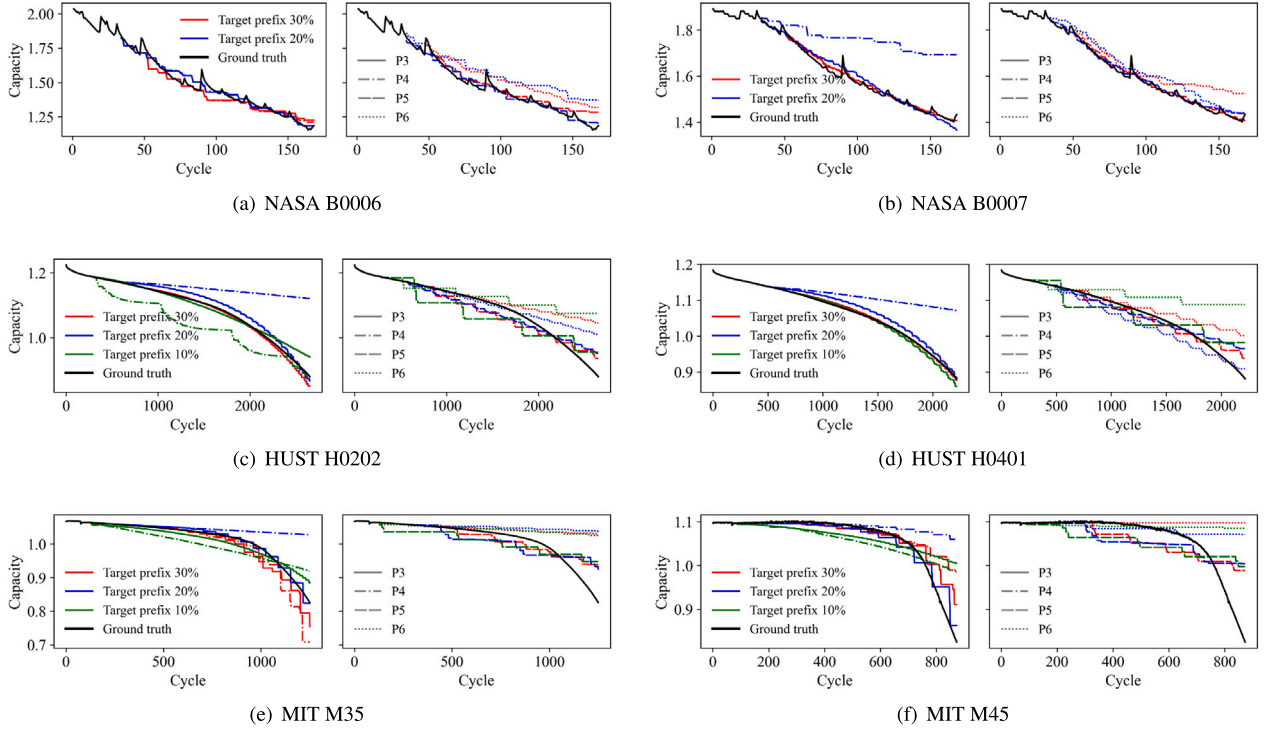


Fig. 10. TimesFM prediction trajectories of the target batteries under different target-prefix ratios and transfer settings.

Table 12

Inference time (s) of Group II baselines across the source-adapted intra-dataset and cross-dataset settings (P3–P6).

Pipeline	Method	B0006	B0007	C36	C37	H0202	H0401	M35	M45
P3	LSTM	0.01	0.00	0.04	0.04	0.03	0.02	0.00	0.01
	GRU	0.01	0.02	0.01	0.01	0.01	0.01	0.01	0.00
	TCN	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01
	Transformer	0.00	0.00	0.01	0.01	0.02	0.01	0.02	0.01
	TimesFM	0.42	0.44	0.40	0.45	1.35	1.16	2.29	1.61
P4	LSTM	0.00	0.00	0.03	0.01	0.05	0.07	0.01	0.01
	GRU	0.01	0.01	0.02	0.02	0.01	0.01	0.00	0.00
	TCN	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01
	Transformer	0.00	0.00	0.01	0.02	0.02	0.02	0.02	0.01
	TimesFM	0.34	0.39	0.39	0.44	0.00	0.00	2.42	1.62
P5	LSTM	0.00	0.00	0.00	0.00	0.07	0.06	0.01	0.01
	GRU	0.01	0.00	0.01	0.01	0.13	0.09	0.06	0.04
	TCN	0.00	0.00	0.01	0.01	0.05	0.04	0.02	0.01
	Transformer	0.00	0.00	0.01	0.01	0.06	0.05	0.06	0.04
	TimesFM	0.22	0.16	0.49	0.50	2.73	2.32	1.30	0.95
P6	LSTM	0.00	0.00	0.00	0.00	0.11	0.04	0.02	0.01
	GRU	0.00	0.01	0.02	0.02	0.11	0.09	0.05	0.04
	TCN	0.00	0.00	0.01	0.01	0.05	0.05	0.02	0.02
	Transformer	0.00	0.00	0.01	0.01	0.06	0.05	0.06	0.05
	TimesFM	0.20	0.10	0.44	0.49	2.76	2.32	1.32	0.92

more sensitive to both prefix ratio and cell identity. For the two HUST cells, P4 of 30% prefix remains the most stable. For the MIT cells, P4 and P6 often exhibit more abrupt error growth within a smaller number of recursive steps, reflecting the shorter rollout span but sharper late-stage divergence. Importantly, in many cases the major error increase occurs near the end of the forecast horizon rather than immediately after the split, suggesting that practical prediction quality around the relevant degradation region may remain acceptable even when the very last recursive steps become less stable.

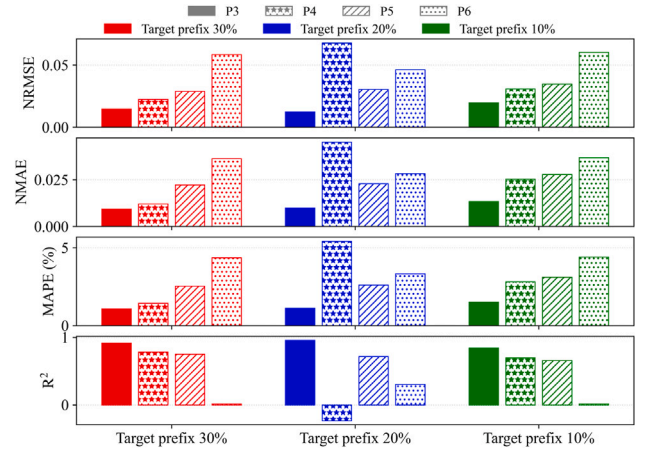


Fig. 11. Sensitivity analysis of TimesFM across target-prefix ratios under the source-adapted settings (P3–P6).

4.4. Fine-tuning dynamics and overfitting analysis

To better understand behavioral differences between a pretrained FM and a smaller baseline during downstream adaptation, we further provide a process-level analysis from two complementary perspectives: (i) fine-tuning dynamics, including gradient-proxy evolution and cross-stage prediction drift; and (ii) overfitting-oriented ablation under limited-prefix supervision. This analysis goes beyond endpoint forecasting accuracy and reveals how adaptation unfolds under data-scarce battery settings. We use HUST H0202 as a representative long-horizon case because it has a relatively long suffix among the batteries included in the sensitivity study, making recursive effects more visible.

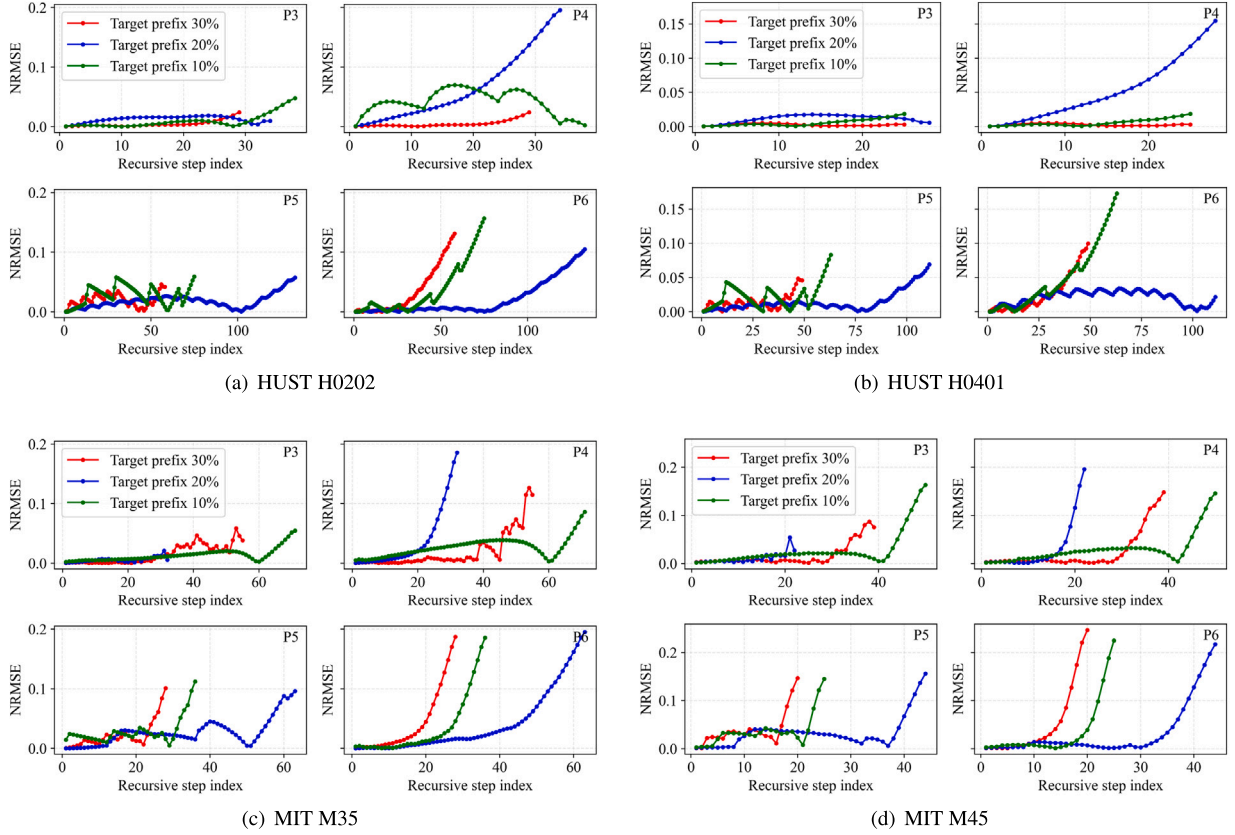


Fig. 12. Step-wise recursive NRMSE of TimesFM on four representative target cells from the HUST and MIT datasets under different target-prefix ratios (30%/20%/10%) and source-adapted transfer settings (P3–P6). The horizontal axis denotes the recursive step index, i.e., the rollout block number during suffix forecasting, and the vertical axis shows the block-wise NRMSE at each recursive step.

Fig. 13 reports the fine-tuning dynamics on HUST H0202. The upper panel shows the gradient proxy, which is calculated in (5), and the lower panel shows cross-stage prediction drift, measured shown in (6).

$$g(e) = \frac{|\mathcal{L}_{\text{train}}(e) - \mathcal{L}_{\text{train}}(e-1)|}{\mathcal{L}_{\text{train}}(1)} \quad (5)$$

where $e \in \{2, \dots, E\}$ is the fine-tuning epoch index, $\mathcal{L}_{\text{train}}(e)$ is the mean training loss at epoch e , and $\mathcal{L}_{\text{train}}(1)$ is the first-epoch training loss used for normalization. Thus, $g(e)$ measures the relative update intensity during adaptation.

$$D_s = \frac{1}{y_0} \sqrt{\frac{1}{N_s} \sum_{i=1}^{N_s} (\hat{y}_{P4,i}^{(s)} - \hat{y}_{P3,i}^{(s)})^2} \quad (6)$$

where, s denotes the recursive-step (block) index, N_s is the number of samples in step s (the last step may be shorter than the nominal step length). $\hat{y}_{P3,i}^{(s)}$ and $\hat{y}_{P4,i}^{(s)}$ are the P3- and P4-stage predictions for the same sample, respectively. Therefore, D_s measures how strongly target-prefix adaptation (P4) changes the forecast relative to the source-adapted target-free baseline (P3) on the same target trajectory.

Several patterns are visible. First, the adaptation dynamics are strongly prefix-dependent. For TimesFM, the gradient proxy under the 30% and 10% prefixes decreases more clearly as training proceeds, indicating progressive stabilization, whereas the 20% prefix exhibits several transient spikes and is visibly less stable. By contrast, the LSTM baseline remains relatively flatter and noisier across epochs, with weaker decay of the gradient proxy. Second, the forecast drift is not uniform across prefixes. Under the 30% prefix, TimesFM produces almost no cross-stage drift, whereas LSTM exhibits a steadily increasing deviation from P3 over the recursive rollout horizon. Under the

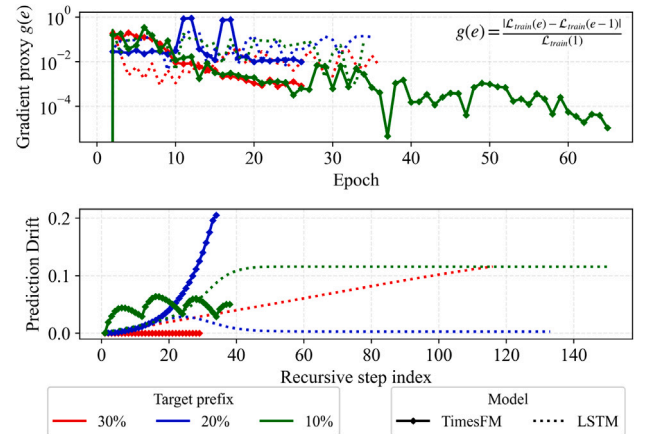


Fig. 13. Gradient-proxy evolution and cross-stage prediction drift (P3–P4) on HUST H0202 under different target-prefix ratios.

10% prefix, TimesFM shows moderate but bounded drift, while LSTM again rises to a substantially higher and more persistent level. The main exception is the 20% prefix, where TimesFM shows the largest forecast drift and eventually exceeds the LSTM baseline. This indicates that TimesFM is not uniformly insensitive to target-prefix adaptation; rather, its stability depends on how informative the observed prefix is.

Fig. 14 further summarizes overfitting-oriented ablation using three indicators: $\Delta\text{NRMSE} = \text{NRMSE}(\text{P4}) - \text{NRMSE}(\text{P3})$, validation worsen (%), and the best-epoch ratio (best epoch/total epochs). These statistics reveal mixed but interpretable differences between TimesFM and

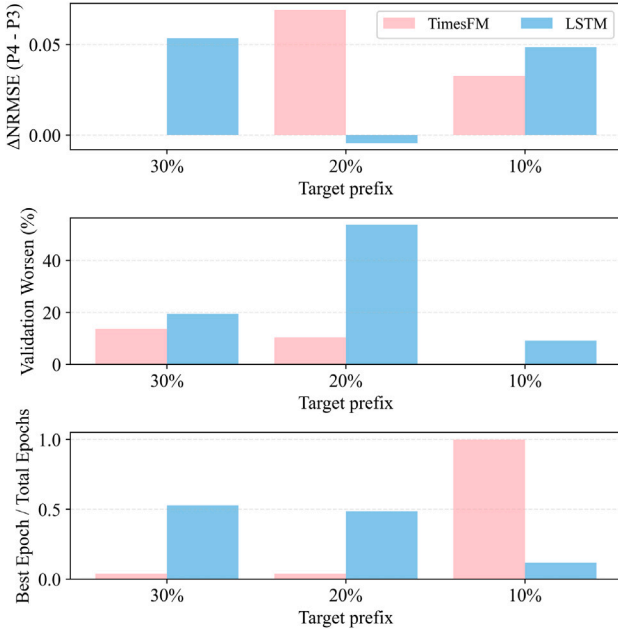


Fig. 14. Overfitting-oriented comparison between TimesFM and LSTM on HUST H0202 under different target-prefix ratios.

LSTM. For the 30% prefix, TimesFM causes almost no increase in forecasting error relative to P3, whereas LSTM shows a clear degradation. For the 10% prefix, both models deteriorate, but the increase remains smaller for TimesFM than for LSTM. The most difficult case is again the 20% prefix, for which TimesFM exhibits the largest ΔNRMSE , while LSTM shows a slight improvement relative to P3.

At the same time, the validation-based indicators still favor TimesFM overall. The validation-worsening percentage is consistently smaller for TimesFM than for LSTM, especially under the 20% prefix, where LSTM shows the largest validation deterioration. Moreover, the best-epoch ratio indicates that TimesFM reaches its best validation state much earlier than LSTM under the 30% and 20% prefixes, whereas LSTM typically requires substantially longer fine-tuning. The 10% prefix is a notable exception, since the best epoch for TimesFM occurs at the end of training, suggesting that under extremely limited supervision it no longer exhibits a clearly early-stopping-dominated regime. Overall, these results suggest that TimesFM is generally less prone to prolonged overfitting and persistent forecast drift than LSTM under the 30% and 10% prefixes, but that the 20% prefix can trigger a distinctly less stable adaptation regime for P4 on H0202.

Overall, compared with battery-specific model-based methods, the proposed framework relies less on explicit physical parameterization but provides weaker direct electrochemical interpretability. Compared with recent task-specific data-driven approaches, it is less specialized to battery-specific supervision or transfer assumptions, but better suited to systematically evaluating transferability under a unified capacity-only, prefix-based setting. Thus, the main contribution of this work is not to claim universal superiority, but to provide a controlled study of when FM-based battery forecasting is effective and where its limitations remain.

5. Conclusion and future work

This work examines cross-dataset generalization of time-series foundation models for lithium-ion batterie remaining useful life forecasting using capacity-only inputs. We establish a unified six-pipeline protocol that factorizes (i) pretrained zero-shot inference, (ii) source-side adaptation on full-life source cells, and (iii) prefix-only target adaptation, spanning both intra-dataset and cross-dataset transfer. Across four

benchmarks, TimesFM shows reliable transfer from short capacity histories, while source-adapted variants yield the strongest and most stable performance, especially under distribution shift and long-horizon recursion. Prefix-only target tuning provides limited gains and can reduce accuracy in some cases. Overall, the results support foundation models pretraining and source-side adaptation as a robust, capacity-only route to remaining useful life prediction with limited target data.

In future work, we will focus on two practical extensions. First, we will study the sample efficiency of TimesFM by varying the available target history to identify the minimal cycle length that still yields accurate long-horizon forecasts. Second, we will enrich the inputs beyond capacity by incorporating voltage, current, and temperature traces, enabling the model to leverage coupled electrochemical dynamics and enhance early end of life detection. These directions aim to make foundation models forecasting more data-efficient and more reliable for real-world battery prognostics.

Another practical direction is to investigate label-noise robustness. In battery prognostics, noise may arise not only from measurement fluctuations in capacity trajectories but also from preprocessing differences and uncertainty in end of life threshold definitions. Extending foundation model based battery forecasting toward noise-aware or label-correction settings is therefore an important topic for future work.

CRedit authorship contribution statement

Lu Zhang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Demba Diallo:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Claude Delpha:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Mohamed Benbouzid:** Writing – review & editing, Visualization, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Expanded intra-dataset validation for P3–P4

This appendix reports supplementary intra-dataset validation results for the P3–P4 pipelines on additional target cells from the HUST and MIT datasets. Capacity trajectory forecasts are shown in Fig. 15, and the corresponding evaluation metrics are reported in Table 13.

Overall, the expanded target-cell results remain qualitatively consistent with the main-text findings: P3 generally provides stable source-adapted forecasting, while P4 can further improve some cases but is more sensitive to the amount of available target prefix. These supplementary results therefore support the robustness of the main conclusions with respect to target-cell selection.

Appendix B. Additional cross-dataset source–target pairing experiments for P5–P6

To further address the concern regarding the rationality of the cross-dataset source–target setup, this appendix provides an additional pairing analysis for the P5–P6 pipelines using HUST and MIT as alternative source/target domains. Specifically, we examine two supplementary transfer cases, namely HUST→MIT and MIT→HUST. The corresponding

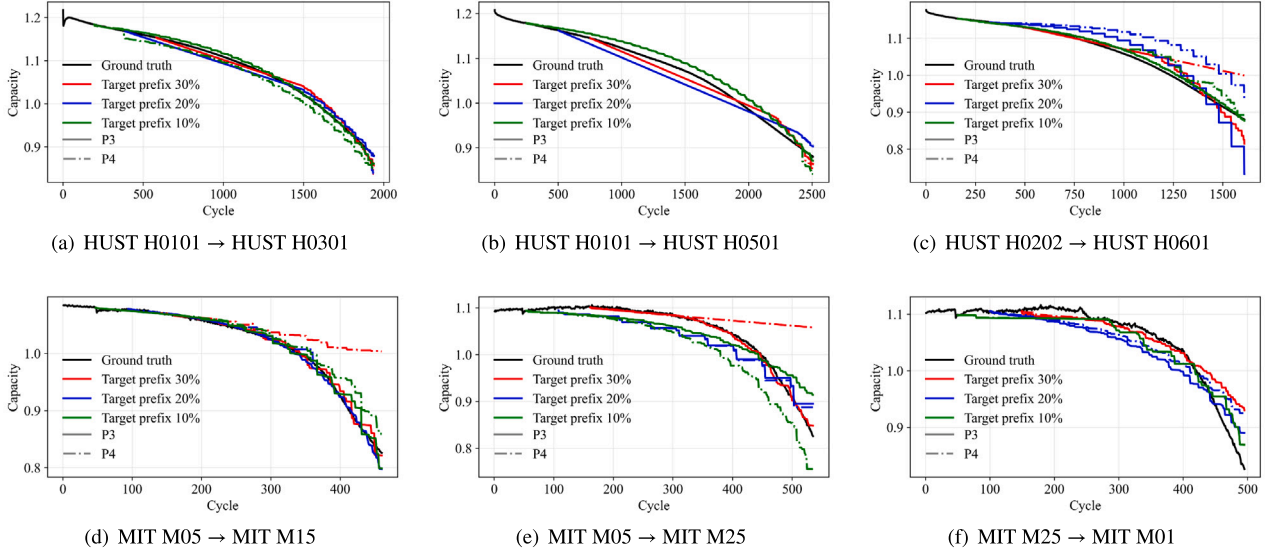


Fig. 15. Expanded target-cell evaluation under the intra-dataset source-adapted settings (P3–P4): capacity trajectory forecasts on additional HUST and MIT batteries.

Table 13

Evaluation metrics for expanded intra-dataset target-cell validation under the source-adapted settings (P3–P4).

Source	Target	Target prefix	Pipeline	Metrics			
				NRMSE	NMAE	MAPE (%)	R ²
HUST H0101	HUST H0301	30%	P3	0.0078	0.0066	0.7858	0.9850
			P4	0.0082	0.0067	0.7982	0.9835
		20%	P3	0.0090	0.0082	0.9292	0.9817
			P4	0.0094	0.0083	0.9547	0.9800
		10%	P3	0.0051	0.0042	0.4783	0.9945
			P4	0.0134	0.0123	1.4387	0.9622
HUST H0101	HUST H0501	30%	P3	0.0109	0.0096	1.1341	0.9715
			P4	0.0113	0.0099	1.1731	0.9695
		20%	P3	0.0174	0.0158	1.8290	0.9348
			P4	0.0175	0.0158	1.8325	0.9345
		10%	P3	0.0127	0.0114	1.3175	0.9687
			P4	0.0137	0.0124	1.4563	0.9631
HUST H0202	HUST H0601	30%	P3	0.0148	0.0108	1.2909	0.9453
			P4	0.0426	0.0286	3.5500	0.5477
		20%	P3	0.0274	0.0222	2.5942	0.8289
			P4	0.0535	0.0454	5.3625	0.3479
		10%	P3	0.0063	0.0053	0.6086	0.9915
			P4	0.0178	0.0127	1.5022	0.9318
MIT M05	MIT M15	30%	P3	0.0066	0.0045	0.5265	0.9896
			P4	0.0593	0.0371	4.4351	0.1669
		20%	P3	0.0053	0.0033	0.3774	0.9934
			P4	0.0064	0.0043	0.4835	0.9904
		10%	P3	0.0078	0.0048	0.5456	0.9853
			P4	0.0179	0.0113	1.3206	0.9226
MIT M05	MIT M25	30%	P3	0.0050	0.0037	0.4057	0.9944
			P4	0.0684	0.0393	4.6234	0.0456
		20%	P3	0.0260	0.0239	2.5235	0.8422
			P4	0.0261	0.0240	2.5228	0.8414
		10%	P3	0.0237	0.0198	2.1369	0.8608
			P4	0.0413	0.0357	3.8538	0.5760
MIT M25	MIT M01	30%	P3	0.0273	0.0173	1.9924	0.8434
			P4	0.0270	0.0166	1.9258	0.8460
		20%	P3	0.0254	0.0228	2.4286	0.8554
			P4	0.0260	0.0209	2.3001	0.8481
		10%	P3	0.0173	0.0143	1.5426	0.9285
			P4	0.0171	0.0141	1.5240	0.9297

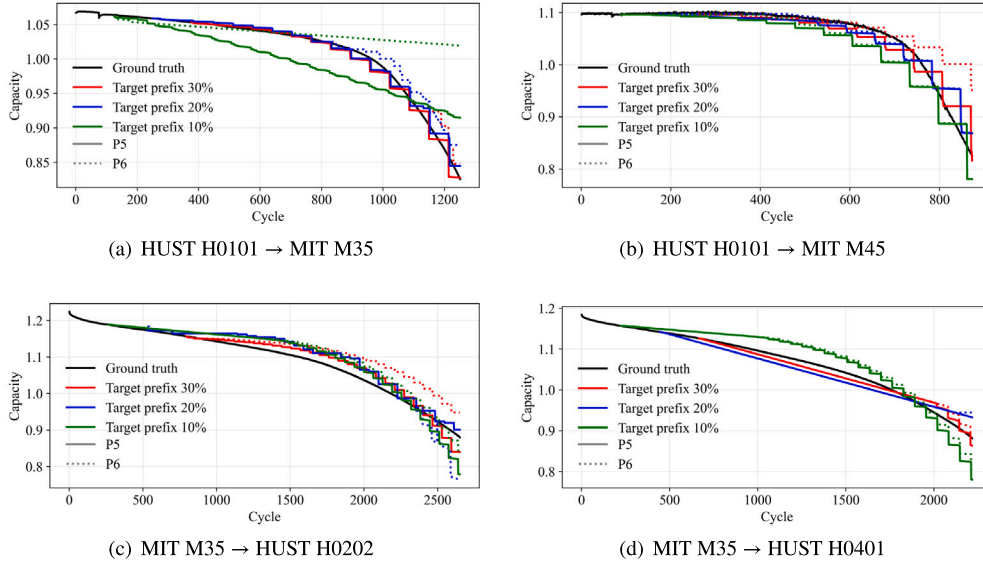


Fig. 16. Expanded target-cell evaluation under the cross-dataset settings (P5–P6): capacity trajectory forecasts on additional target cells.

Table 14

Evaluation metrics for expanded cross-dataset target-cell validation under the source-adapted settings (P5–P6).

Source	Target	Target prefix	Pipeline	Metrics			
				NRMSE	NMAE	MAPE (%)	R ²
HUST H0101	MIT M35	30%	P3	0.0074	0.0050	0.5634	0.9823
			P4	0.0110	0.0066	0.7579	0.9615
		20%	P3	0.0070	0.0049	0.5430	0.9840
			P4	0.0129	0.0083	0.9478	0.9457
		10%	P3	0.0300	0.0255	2.7539	0.6993
			P4	0.0524	0.0283	3.2891	0.0825
HUST H0101	MIT M45	30%	P3	0.0189	0.0127	1.4225	0.9151
			P4	0.0417	0.0204	2.4274	0.5853
		20%	P3	0.0168	0.0118	1.2957	0.9276
			P4	0.0158	0.0087	0.9848	0.9356
		10%	P3	0.0236	0.0183	1.9456	0.8452
			P4	0.0217	0.0149	1.5962	0.8700
MIT M35	HUST H0202	30%	P3	0.0175	0.0151	1.7952	0.9251
			P4	0.0388	0.0333	4.0280	0.6329
		20%	P3	0.0215	0.0187	2.1314	0.8957
			P4	0.0298	0.0240	2.8517	0.7993
		10%	P3	0.0225	0.0186	2.1749	0.8929
			P4	0.0231	0.0200	2.2993	0.8868
MIT M35	HUST H0401	30%	P3	0.0098	0.0086	0.9983	0.9698
			P4	0.0116	0.0098	1.1576	0.9579
		20%	P3	0.0168	0.0148	1.7066	0.9216
			P4	0.0174	0.0151	1.7493	0.9152
		10%	P3	0.0246	0.0206	2.3712	0.8456
			P4	0.0239	0.0206	2.3496	0.8547

trajectory forecasts and quantitative evaluation metrics are reported in Fig. 16 and Table 14, respectively.

Compared with the main-text setting in which NASA and CALCE are jointly used as the source domain, these HUST/MIT pairings generally achieve better forecasting performance. A plausible reason is that NASA exhibits relatively more linear degradation trajectories, whereas CALCE contains stronger local fluctuations, and both datasets have shorter life spans than HUST and MIT in most cases. Nevertheless, even under the more mismatched NASA+CALCE source setting used in the main text, the results still demonstrate that TimesFM maintains competitive temporal forecasting capability under challenging transfer conditions.

Data availability

Data will be made available on request.

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